

BitVoice manual

An open-source phone system in one program, set up entirely in the web interface: extensions, queues, conference rooms, the staff portal, hosted organisations, and the public-safety lab track.

63 chapters · 63 figures · 20 thousand words · BioDream LLC

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PART 1

Getting started

From a new install to a working phone system: extensions, phones, a phone company, where calls go, emergency calling from your offices, and how to tell everything is ready.

CHAPTER 1

Getting started

In the web interface: `/`

BitVoice is a complete phone system in one program: the exchange that phones sign in to, the menus and queues callers move through, voicemail, recording, conference rooms, faxes, and the connections to phone companies. Everything is set up in the browser, on the screens this manual describes. There is no configuration file to edit and nothing else to install.

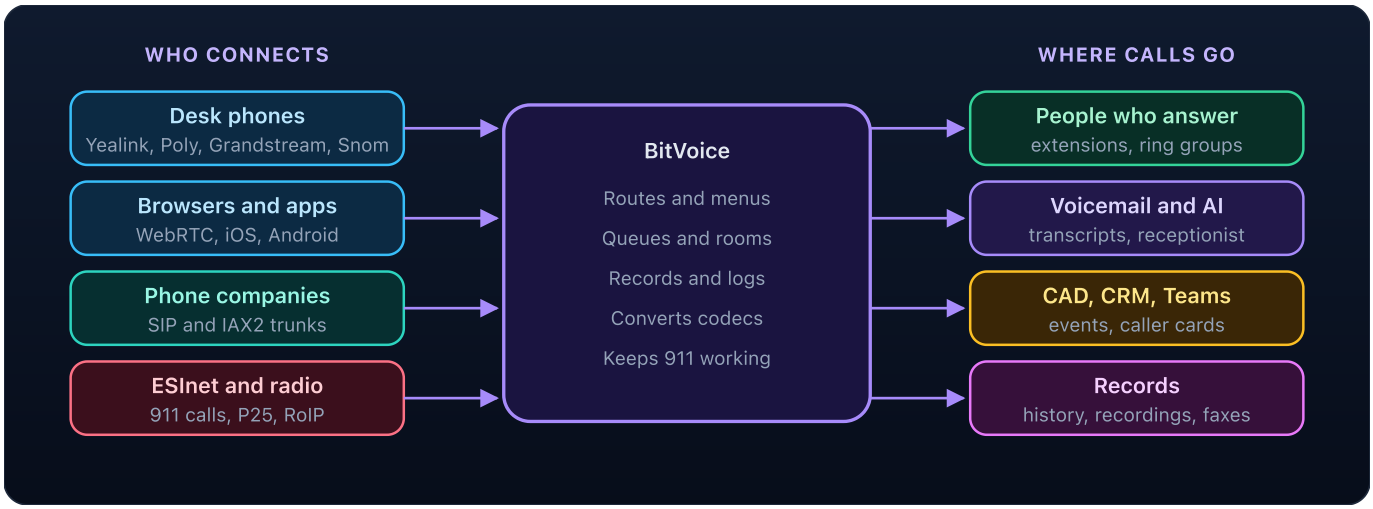


Figure 1. Where BitVoice sits: phones and carriers on one side, people and systems on the other

What you need

- **A computer to run it on**: a small office server, a virtual machine, a cloud instance, or for a first look your own laptop. Linux, macOS and Windows all work. A Raspberry Pi 4 runs a small office.

- **Phones**: desk phones from Yealink, Poly, Grandstream or Snom are set up automatically; any other SIP phone or softphone works with its own settings; and every browser is a phone too.
- **A trunk** (an account with a phone company) to call outside numbers and be called. You can start without one: extensions call each other and the echo test.

The first start

Start BitVoice and open the address it prints (on a desktop it opens the browser itself). The **setup wizard** asks four things: the organisation's name and country, the first administrator's password, the first extensions, and your phone company if you have one. The country sets the emergency numbers, the tones callers hear and the way phone numbers are written.

> Choose the country carefully. It decides which numbers are emergency numbers (911 and 933 in the US, 112 and 999 elsewhere), and emergency numbers always work from every extension.

The first hour

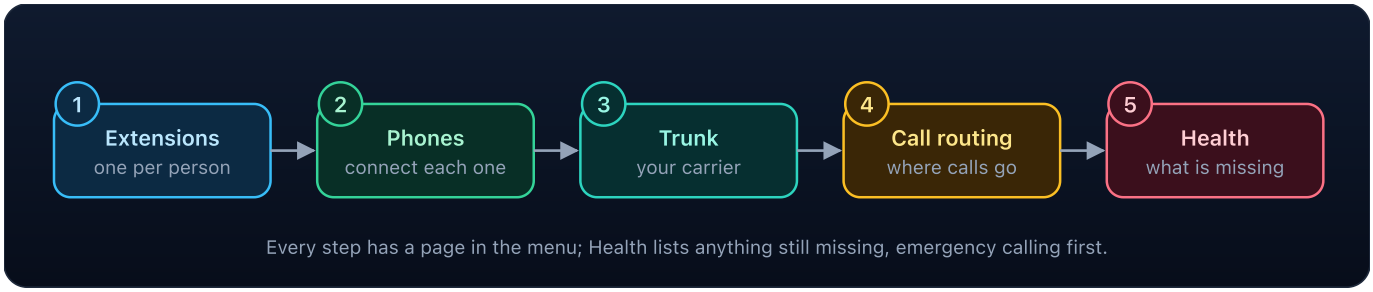


Figure 2. The first hour: five steps from a new install to a working phone system

1. **Extensions**: one per person or desk. Each has a number and a SIP password. Open Extensions (chapter 2)
2. **Phones**: connect a desk phone, softphone or the browser phone to each extension. Open Phones (chapter 3)
3. **Trunk**: your phone company account, so you can call outside numbers and be called. Open Trunks (chapter 4)
4. **Call routing**: where calls from outside go (a person, a ring group, a menu), and which trunk outside calls use. Open Call routing (chapter 5)
5. **Health** tells you what is still missing, emergency calling first. Open Health (chapter 8)

Then make a test call: from any phone dial `*43`, the echo test. You hear yourself back, which proves sound works both ways.

Where it runs

BitVoice runs on Linux, macOS and Windows.

System	How to start it	Where its data is
Linux	`./bitvoice` (or the systemd service in the release)	`~/.config/BitVoice`
macOS	`./bitvoice`	`~/Library/Application Support/BitVoice`
Windows	double-click `bitvoice.exe`	`%APPDATA%\BitVoice`

On Windows, keep the window it opens while you use BitVoice (closing it stops BitVoice), and when Windows asks whether it may use the network, allow private networks so phones can connect. `-data` puts the data anywhere else; `-http`, `-https` and `-sip` choose the addresses and ports.

The ports it uses

Port	What for
5060 UDP and TCP	SIP: phones and trunks
5061 TCP	SIP over TLS (encrypted signalling)
4569 UDP	IAX2 trunks, only once one is set up
10000–20000 UDP	the sound of calls (RTP), set in Settings
8280 TCP (this computer only), 8443 TCP	this web interface: plain on this computer, HTTPS from others

If another phone system already has 5060, BitVoice takes the next free port and Health says so.

Everything in the web interface

Every setting has a screen, and every screen has a chapter here. Nothing needs a configuration file, a command line or a second tool. For automation, the same screens are a REST API, and the configuration can follow a file in Git (Configuration as code (chapter 54)), but neither is ever required.

What BitVoice is for

BitVoice is first a phone system for offices and contact centres, and a hosted phone service for providers: everything in the first five parts of this manual. The 911 answering-point profile, the radio dispatch console and NG911 Core Services are a separate **lab and design-partner track** (Public safety (chapter 61)): they are off the menu until switched on in Settings, and not certified for live use. Emergency calls from your own phones are not part of that track; they work on every phone system from the first start.

Finding your way

- Press **⌘K** (Ctrl+K) anywhere to jump to a screen, or type an extension's name or number to open it.
- Every screen has a **Help** link at the top right that opens its chapter of this manual.
- **Changes** keeps every version of the configuration: you can see what changed and undo it.
- Three themes (Midnight, Dark and Light) are at the foot of the menu.

CHAPTER 2

Extensions

In the web interface: `/extensions`

An extension is a number for a person or a desk. Phones sign in with the extension number and its **SIP password**; calls to the number ring every phone signed in as it, at once.

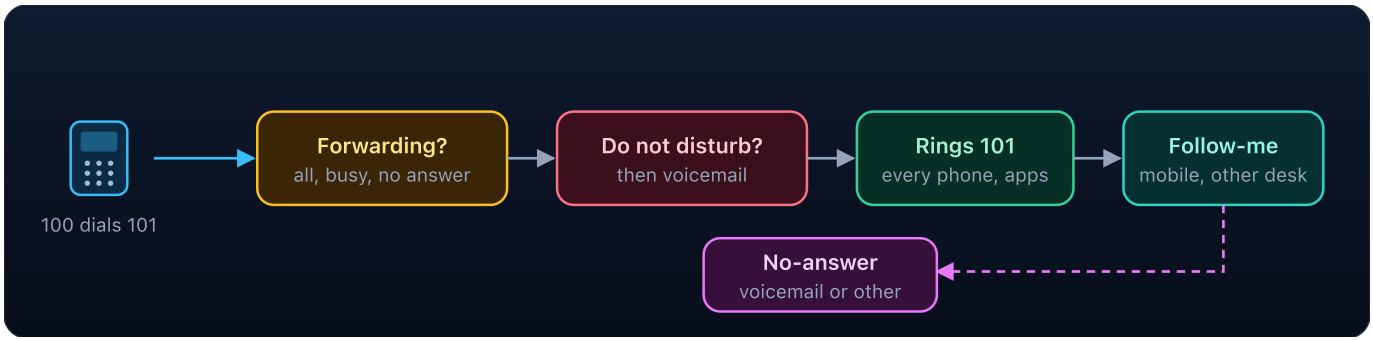


Figure 3. A call between two extensions

Adding an extension

1. Open Extensions (chapter 2) and press **Add extension**. The next free number and a strong SIP password are filled in.
2. Enter the person's **name** (shown on other phones when they call) and **email** (for voicemail by email).
3. Fill in the **911 location**: the site, floor and room. It is sent with emergency calls.

4. Choose what happens **If nobody answers**: voicemail by default, or another extension, a group or a menu.
5. Save. The extension can sign in at once.

The fields

Field	What it does
Number	2 to 8 digits; no other extension, group, queue, room or feature code may have it
SIP password	what the phone signs in with; at least 12 characters; **Generate** makes a new one
Ring for	seconds before **If nobody answers** happens (25 by default)
Outside caller ID	the number shown on calls out; empty uses the trunk's
Can call outside numbers	off keeps the phone to internal calls and emergency numbers
Record calls	when Settings records only selected extensions
Hot desk	a shared phone that becomes whoever signs in at it
Encrypt calls to this phone	media encryption (SRTP); use it with TLS so the keys stay private
Department	groups this extension's call costs

> Fill in the 911 location for every desk. An emergency call carries it to the answering point, and the RAY BAUM'S Act requires a dispatchable location for calls from offices in the US.

Follow-me, pickup and intercom

Each person's own call handling is in its own box on the extension:

- **Follow-me** rings more numbers when the phones here are not answered: a mobile, another desk. They ring one after another, or all together, for their own ring time. With **Outside numbers press 1 to take the call** on, whoever answers on a mobile presses 1 first, so the mobile's own voicemail never takes the call. See Feature codes (chapter 20).
- **Pickup group**: anyone in the same group answers this extension's ringing calls by dialling `*8`.
- **Allow intercom**: `*80` and this number answers on the speaker by itself. Switch it off for people who never want that.
- **Own speed dials**: short codes for this person only, dialled after `*0`.

Forwarding right now

Below the settings, **Forwarding right now** shows and changes what the person set from their phone or the staff portal: do not disturb, and forwarding of every call, of calls while busy, and of calls not answered. These are live state, not configuration: changing them makes no revision.

What is connected

- **Phone**: a desk phone, softphone or app.
- **Door station**: an intercom at a door. Whoever answers presses the key code, or **Open door** on Live calls. With a door controller address, BitVoice calls that address (2N, Axis, Akuvox, relay boards); without one, the code is sent to the station, which opens the door itself.
- **Overhead speaker**: calls to it are answered at once and heard over the speaker.
- **Analog adapter port**: an old phone, fax or alarm panel on an adapter. Calls use G.711 only, which faxes need.

Connected devices

An extension that is open shows every device signed in as it: the phone's make and model, how it connects (UDP, TCP, TLS or the browser) and since when. The address and port to give a phone are shown there too.

CHAPTER 3

Connecting phones

In the web interface: `/phones`

Any SIP phone or app works: point it at this server's address and port 5060, with the extension number as user and its SIP password. Desk phones from the four big makers are set up automatically instead, so nobody types a password into a phone's keypad.

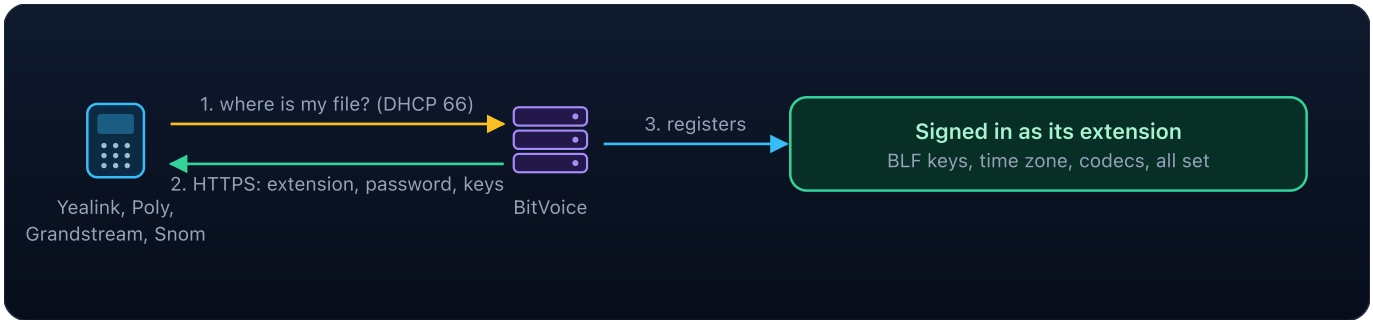


Figure 4. Provisioning: a desk phone fetches its settings by itself

Automatic set-up (provisioning)

1. On Phones (chapter 3) press **Add phone** and enter the phone's **MAC address** (on a sticker underneath) and the **extension** it belongs to.
2. Choose the make: Yealink, Poly, Grandstream (including the HT analog adapters) or Snom.
3. Tell the phone where BitVoice is: in your DHCP server, set **option 66** to the provisioning address shown on the screen, so every phone on the network finds it by itself; or type the same address in the phone's own web page, under auto provisioning.
4. Restart the phone. It fetches its file over HTTPS, signs in, and its line keys show who is on the phone.

The file carries the extension, its password, the time zone, the codecs, busy lamp keys and the address of this server. Provisioning is protected by a user name and password the phones are given once.

Make	Models tested	What it gets
Yealink	T3x, T4x, T5x	account, BLF keys, time, codecs
Poly	VVX, Edge E	the same, in Poly's own format
Grandstream	GRP, GXP, HT802/814	account per port on adapters
Snom	D7xx	account, keys, time

Phones without provisioning

Softphones and other makes take the same three things: server (this computer's address, port 5060, or 5061 with TLS), user (the extension number) and password (its SIP password). Health lists phones that tried a wrong password; after a few tries the address is banned for a while (see Security (chapter 45)).

The browser phone

The phone icon at the bottom of every screen makes this browser a phone for an extension, with your microphone and speakers. It uses WebRTC: the sound is encrypted end to end, and it works through most firewalls without any setting.

The staff phone

Open `#/phone` on any computer or mobile browser: people sign in with their extension and SIP password, without an administrator account. It is a full phone: dial, answer, hold, transfer and the keypad. For their own forwarding and voicemail they use the staff portal at `#/me` (see The staff portal (chapter 21)).

The mobile apps

The BitVoice apps for iPhone and Android are paired from the extension's page with a QR code. Calls wake the app through Apple's and Google's push services, so it rings even when it is closed.

CHAPTER 4

Trunks

In the web interface: `/trunks`

A trunk is your account with a phone company. It carries calls to and from outside numbers. Most are SIP trunks; an Asterisk or FreePBX server, or an IAX2 provider, can be reached with IAX2 instead.

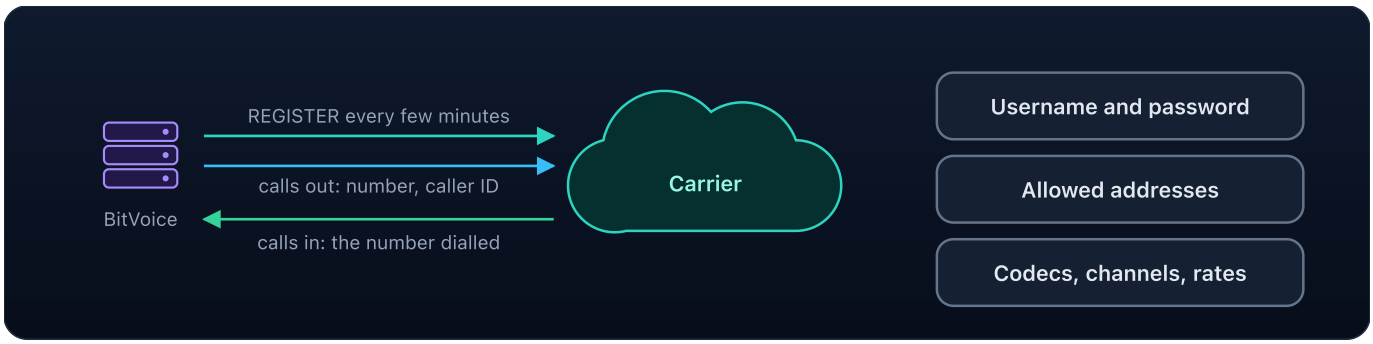


Figure 5. A trunk: your account with a phone company

Adding a SIP trunk

1. On Trunks (chapter 4) press **Add trunk**.
2. Enter the **provider server** your phone company gave you (`sip.provider.com`), and the port if it is not 5060.
3. Enter the **username** and **password**, and switch on **Register** if the provider says to (most do).
4. Save. Within a few seconds the trunk shows **registered**; if not, the reason is shown beside it.
5. Add an **incoming route** for your number and an **outgoing route** that uses this trunk (Call routing (chapter 5)).

Field	What it does
Transport	UDP (usual), TCP, or TLS for encrypted signalling
Register	sign in every few minutes, so the provider knows where to send calls
Authentication username	only when the provider gives a separate one
From domain	the domain in the caller's address, when the provider asks for one
Caller ID	the number shown on calls out; an extension's own caller ID wins
Also accept calls from	the provider's other addresses or networks that send calls in
Calls at once	the limit your plan allows; calls beyond it go to the next trunk in the route
Encrypt media	SRTP, for providers that support it
Codecs	leave it on *same as Settings*, or choose what this provider supports, best first
Rates	what calls cost by number prefix, for Costs (chapter 40)

> A trunk that accepts calls by address alone should list only the provider's own addresses. Networks wider than a /16 are refused: they would let anyone on the internet pass as your provider.

Cloud providers

Start from fills in Microsoft Teams Direct Routing, Amazon Chime SDK Voice Connector or Azure Communication Services, with a note on what to set on their side. Teams is described in its own chapter (Microsoft Teams (chapter 52)).

IAX2 trunks to Asterisk and FreePBX

Choose **IAX2** as the protocol to join another Asterisk or FreePBX server, or an IAX2 provider. IAX2 carries signalling and sound on one UDP port, 4569, which makes it easy through firewalls.

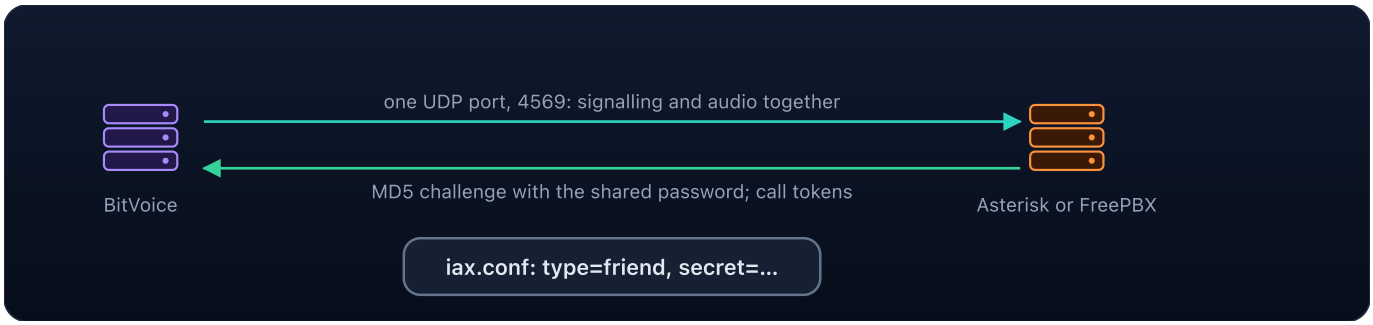


Figure 6. An IAX2 trunk to an Asterisk or FreePBX server

- **Server address**: the other server. Leave it empty when the other server registers to BitVoice instead (a branch office behind a changing address); then the username and password are what it registers with.
- **Username and password**: the same on both sides. BitVoice answers and sends MD5 challenges, and handles call tokens.
- **Register**: sign in to the other server, so it can send calls here.

On the Asterisk side, a peer like this is all it takes:

```
[bitvoice]
type=friend
host=dynamic
secret=the shared password
context=from-bitvoice
```

IAX2 port 4569 opens only when the first IAX2 trunk is added, so an Asterisk running on the same computer keeps it until then.

What the status means

Status	Meaning
registered	signed in; calls can come in
ready	does not register, or IAX2 waiting for calls
unreachable	the server's name is not in DNS, or nothing answers
failed	the provider refused the sign-in; the reason is shown
standby	servers share this trunk one at a time and another holds it

CHAPTER 5

Call routing

In the web interface: `/routes`

Call routing decides two things: where calls from outside go, by the number they dialled, and which trunk calls to outside numbers use.

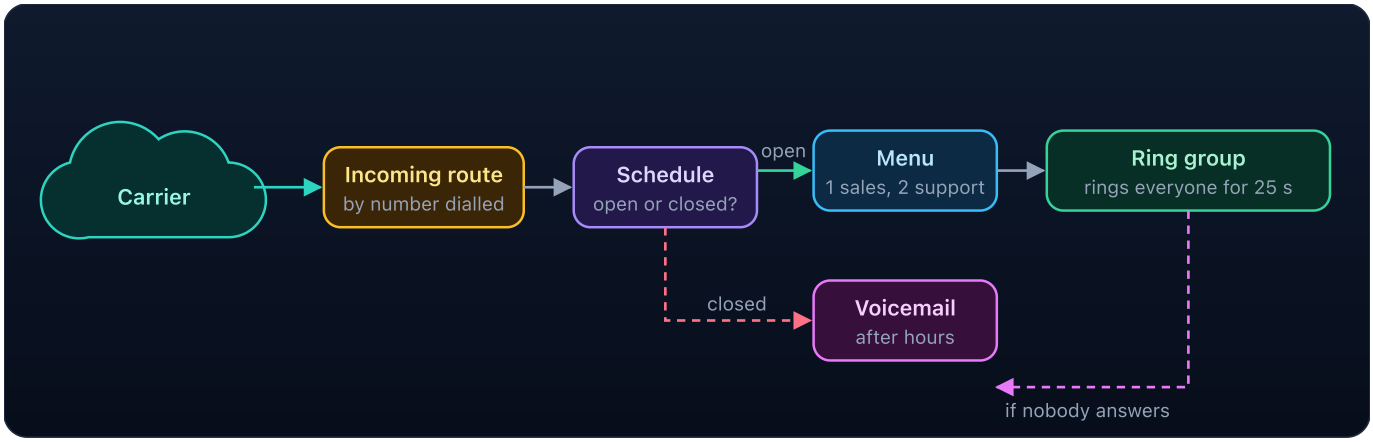


Figure 7. How a call from outside finds its way

Incoming routes

Each incoming route matches the number that was called (or any number) and sends the call to a destination: an extension, ring group, queue, menu, schedule, script, conference room, AI receptionist or voicemail. Numbers match in any form: +1 212 555 0100, 2125550100 and 12125550100 are the same.

- A route for one **number** wins over a route for any number.
- A route can be limited to one **trunk**, for numbers that two providers both send.
- With **emergency mode** on (Settings), every incoming call goes to one place instead, whatever the routes say: an evacuation, a closure, an incident.

Outgoing routes

Outgoing routes match what people dial, by pattern, and send it to trunks in order: if the first is down or full, the next is tried.

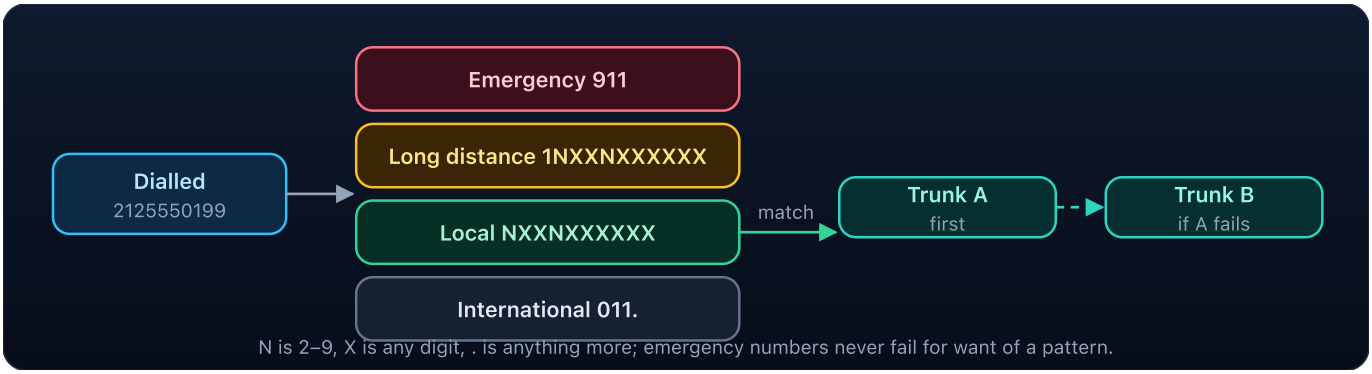


Figure 8. Outgoing routes: the first pattern that matches, its trunks in order

Pattern	Matches
`X`	any digit
`Z`	1 to 9
`N`	2 to 9
`[1-5]`	one of a set
`.` at the end	one or more further digits
`NXXNXXXXXX`	a ten-digit North American number
`1NXXNXXXXXX`	the same with 1 in front
`011.`	an international call from the US

- **Remove** takes digits off the front (a 9 dialled for an outside line); **Then add in front** puts some on (a country code).
- The order matters: the first route whose pattern matches is used. Drag routes to change it.
- Premium numbers (900 and the like) are refused unless a route names them.

Emergency numbers

The emergency numbers always work. A route that names the number exactly is used first; with none, every working trunk is tried in turn. A call to them is never refused for an extension's limits, a hosted organisation's limits or a missing route. Health warns when no trunk can carry them.

> Test emergency routing with your provider's test number (933 in the US) rather than 911. 933 reads back the address the call carries.

CHAPTER 6

Emergency calls

Emergency numbers always go out, from every extension, even ones that cannot call outside numbers. This chapter is about emergency calls made from the phones BitVoice runs: an office calling 911. Answering 911 calls is the 911 profile (chapter 61).

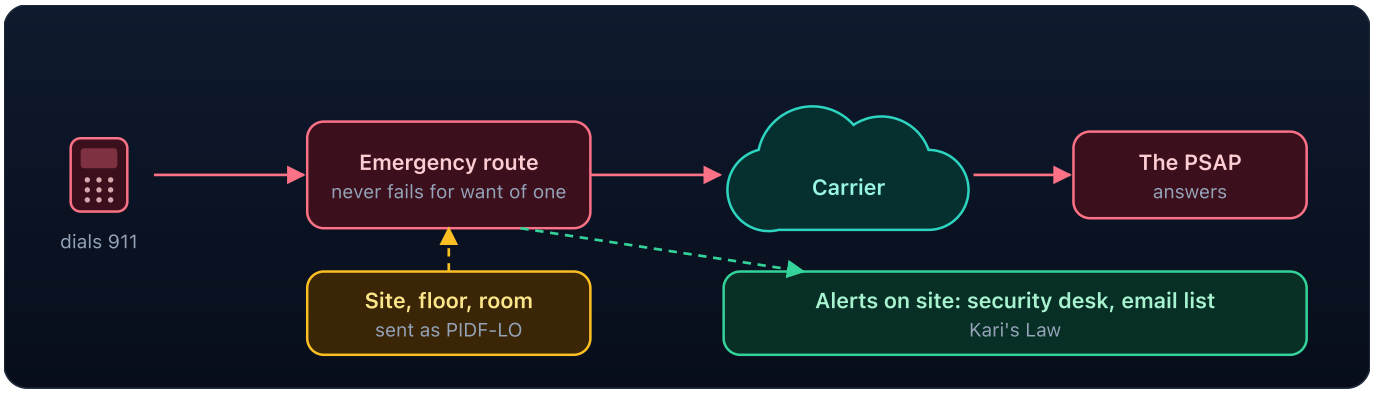


Figure 9. Office 911: location goes with the call, people on site are told

What BitVoice does on every emergency call

1. **Routes it**, always: a route that names the number first; with none, every working trunk in turn. Limits never stop it: an extension not allowed outside calls, a hosted organisation at its call limit or suspended, a trunk at its channel limit.
2. **Sends where the caller is**: with **Send the caller's address** on (Settings), the call carries the site's civic address, floor and room to the provider as PIDF-LO.
3. **Tells people on site** (Kari's Law in the US): the extensions under **Notify** get a message on their phones, every open BitVoice screen shows an alert, and the addresses under **Also email** get an email with the caller's name, extension, site, floor and room, so someone can meet the responders at the door.
4. **Lets the call back through**: for an hour after an emergency call, calls coming in are never refused, so the answering point can always ring back.

Setting it up

1. Check the emergency numbers in Settings (chapter 9) (set by country: 911 and 933 in the US).
2. Add every building under Sites (chapter 7), with its street address and the networks its phones use.
3. Give every extension its site, floor and room.
4. Make sure an outgoing route and trunk carry emergency calls; Health (chapter 8) warns when none can.
5. Fill in **Notify** and **Also email**.

6. Dial `933` (in the US) to hear the address your provider has.

Remote workers

A phone registering from outside its site's networks is a remote worker: their phone page asks where they are, and that address is used instead. Alerts say when it was not confirmed.

Emergency mode

A different thing: one switch that sends every incoming call to one place (an evacuation, a closure, an incident). Set where calls go, an optional code such as `*55` and the extensions allowed to dial it in Settings (chapter 9); switch it on the Dashboard (chapter 1), or by dialling the code (rising tones for on, a low tone for off). A red bar shows on every screen while it is on.

CHAPTER 7

Sites and office 911

In the web interface: `/sites`

A site is a building: its street address and the networks its phones use. Sites are how an emergency call says where the caller is, down to the floor and room. In the US, Kari's Law and RAY BAUM'S Act require exactly this from business phone systems.

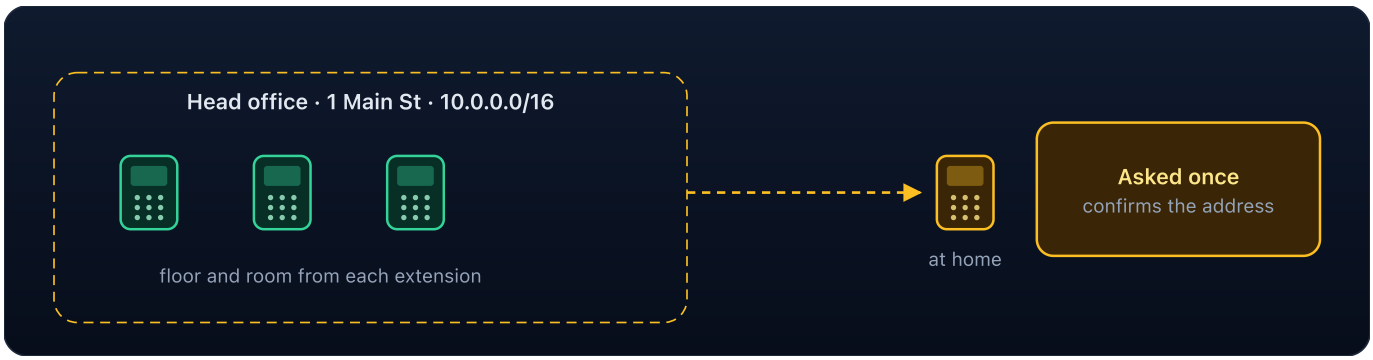


Figure 10. Sites and remote workers: where a phone is, for 911

Adding a site

On Sites (chapter 7), press **Add**:

Field	What it is
Name	what people call it: Head office
Street, City, State or province, Postal code, Country	the civic address responders are sent to
Networks	the address ranges its phones use, such as `10.1.0.0/16`; one per line

Then on each extension give its site, **floor** and **room**. Health (chapter 8) lists extensions without a location.

How a phone's place is found

1. A phone registering from one of a site's networks is at that site, with its extension's floor and room.
2. A hot desk carries the desk's own site, floor and room, not the person signed in (Hot desking (chapter 24)).
3. A phone registering from outside every site's networks is a **remote worker**. On its first emergency-capable registration from a new place, its user is asked to confirm their address on their phone page; emergency calls then carry that address.
4. If a remote worker never confirmed, the call still goes out, with the extension's usual site, and alerts say the address was not confirmed.

On-site alerts

With **Email when someone calls 911** set in Settings, the people named there get the caller's name, extension, site, floor and room as soon as a call to an emergency number starts, so someone can meet the responders at the door. **Notify** extensions get a message on their phones, and every open BitVoice screen shows an alert. See Emergency calls (chapter 6).

> Check each site's address with your provider: dial `933` (in the US) from a phone at the site to hear the address the provider has.

CHAPTER 8

Health

In the web interface: `/health`

Health says whether the system is ready for calls, and if not, what to do. Each line says what is wrong in plain words and links to the screen where it is fixed. Check it after every change of any size.

What it checks

Check	Why it matters
Emergency calls can get out	a route and a working trunk carry the emergency numbers
Extensions without an emergency location	their 911 calls would carry no floor or room
The phone service	SIP is listening, and on which port
Phones not connected	extensions nobody has signed in as
Trunks	registered, or the reason they are not
Menus without a recorded prompt	callers would hear silence
The time zone	schedules follow it
Recordings and their keeping	disk space, and how long recordings are kept
Certificates	when the TLS certificate runs out
Key vaults	secrets a vault cannot read
Other servers	when several share a database, whether each is running

Problems are coloured: red stops calls (no emergency route), amber should be looked at (a phone not connected), and grey is for information.

> The emergency checks are always first. A phone system that cannot call 911 is not ready, whatever else works.

CHAPTER 9

Settings

In the web interface: `/settings`

Settings holds everything about the system as a whole. Each section saves on its own, and every save is a revision on Changes (chapter 46).

Section	What is there
System	name, country, time zone, language, the public address phones and providers see
Numbers	the echo test, voicemail and park numbers, and the emergency numbers
Codecs	the codecs phones may use, best first (Codecs (chapter 51))
Call recording	off, every call, or selected extensions; consent rules; how long recordings are kept
Sound	clean-up for voicemail and during calls
Protection	bans after failed sign-ins, trusted networks, calls per extension, premium numbers
Emergency	send the caller's address, who is notified, who is emailed, emergency mode
AI	the AI service, models, and which AI features are on (AI (chapter 16))
Single sign-on	the identity provider (Single sign-on (chapter 44))
Asterisk compatibility	AMI and ARI for programs written for Asterisk (Asterisk compatibility (chapter 55))
Mobile apps	the push service that wakes the apps
Email	the mail server voicemail, faxes and alerts go through
Key vaults	secrets kept in AWS Secrets Manager or Azure Key Vault
Configuration from Git	keep the configuration in step with a file in a repository
Database	SQLite or PostgreSQL (Database (chapter 49))

Passwords and keys are never shown again once saved; leave them empty to keep them. Viewers see Settings with every secret removed.

> The public address matters when BitVoice is behind a router: phones and providers outside send sound to the address it advertises. Leave it empty to work it out per phone.

PART 2

Where calls go

The building blocks callers move through: groups and queues, menus and schedules, scripts and the AI receptionist, voicemail, and the diagram that shows it all.

CHAPTER 10

Call flow

In the web interface: `/flow`

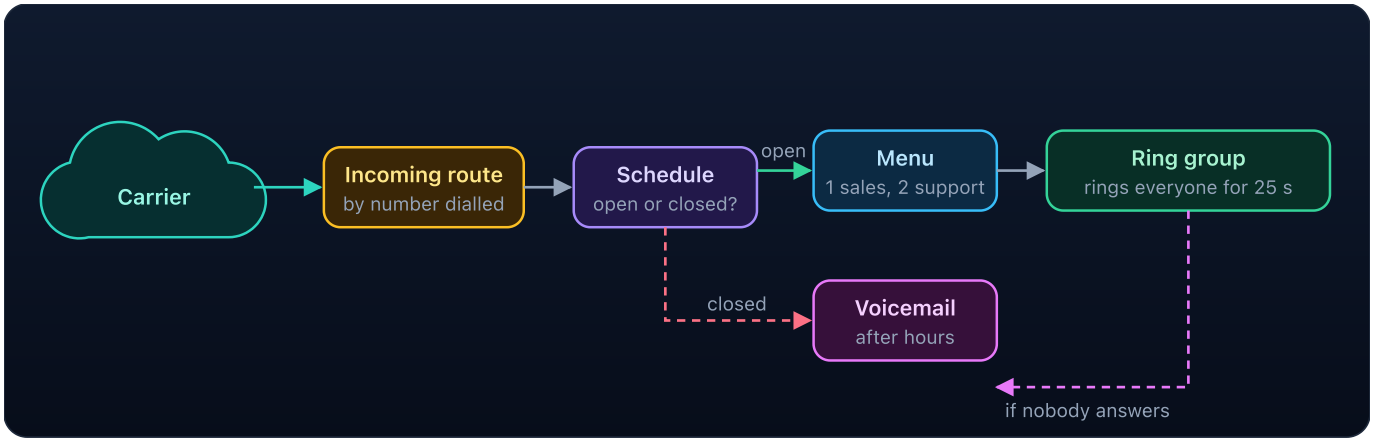


Figure 11. How a call from outside finds its way

Try a call

Enter who calls (an extension, or an outside number), what they dial and when, and press **Try it**. BitVoice follows the call exactly as it would for real, without ringing anything: the incoming route, schedules at that time, routing scripts (they run, in their sandbox), menus, groups and queues, and what happens if nobody answers. The path this call takes is marked; other ways on (a menu's keys, a schedule's other side) can be opened. Green means it works, amber means look at it (a phone not connected, a script falling back), red means the call fails there. Tick **Assume every phone is connected** to see the plan as designed rather than as it is right now; **Arrives on** picks the trunk for routes that only take one.

The diagram

The diagram shows where calls from outside go, box by box: each incoming route, then the menus, groups, queues, schedules, scripts and people it leads to.

Click a box (or Tab to it and press Enter) to change where calls go from it: a route's destination, an extension's or group's **If nobody answers**, a schedule's open and closed destinations, a menu's keys (add or remove keys right there), a script's fallback or a receptionist's person. Saving checks the change like any other and makes a revision. **Open its full editor** goes to the item's own page. **Add incoming route** adds a route without leaving the diagram.

CHAPTER 11

Ring groups

In the web interface: `/ring-groups`

A ring group is one number that rings several extensions: a sales team, the front desk and its back-up, everyone in the workshop.

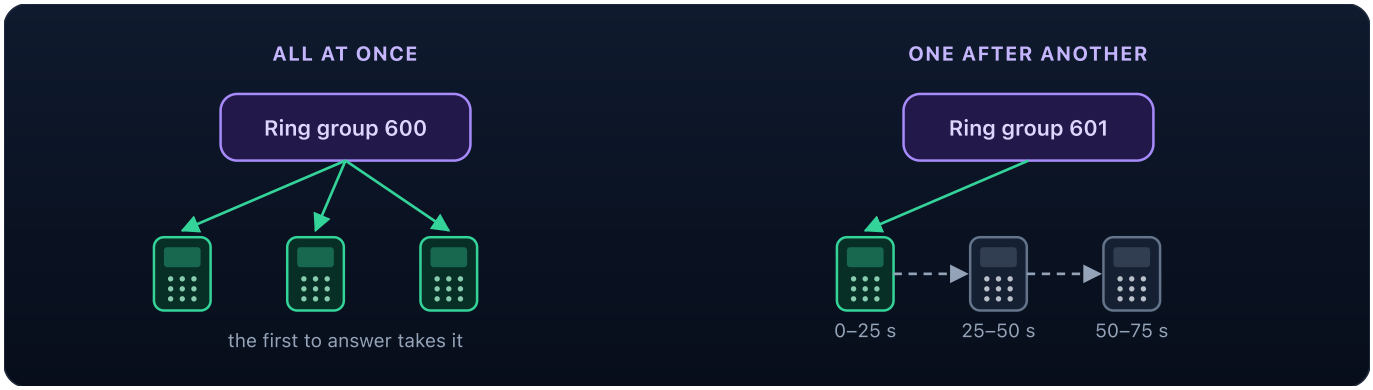


Figure 12. Ring groups: all at once, or one after another

Ringing strategies

- **All at once**: every member's phones ring together, and the first to answer takes the call. The others stop ringing.
- **One after another**: each member rings for the group's ring time, in the order listed. Move members with the arrows, or Alt+↑ and Alt+↓.

Members who are away (in a meeting on their calendar, or with do not disturb on) are skipped; the group rings the others. Members with the mobile app are woken by a push and rung too.

If nobody answers

When nobody answers within the ring time, the call goes to **If nobody answers**: a member's voicemail, a queue, a menu, another group. With nothing set, it rings until the caller gives up.

Groups, queues or pickup?

Use	When
A ring group	a few people, any of whom can take the call, and the call should ring now
A queue	callers may have to wait for someone free, and should hear their place
A pickup group	each person has their own calls, but colleagues may answer for them

---

CHAPTER 12

Queues

In the web interface: `/queues`

A queue holds callers in order, with music, and rings the next free agent. It is the heart of a contact centre, and useful in any office where calls come faster than people can take them.

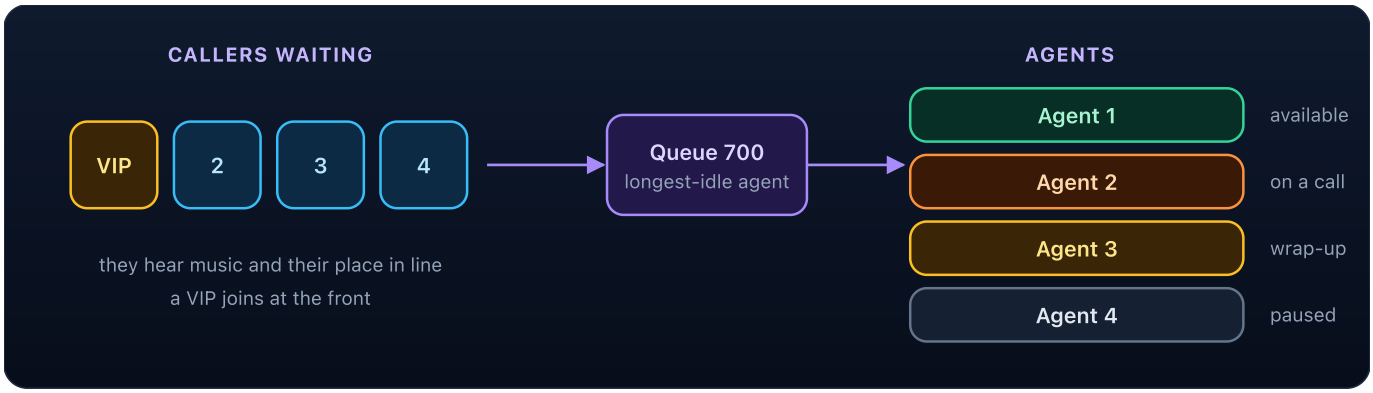


Figure 13. A queue: callers wait in order, agents take the next one

How calls are offered

Strategy	Who is rung
Longest idle	the free agent who has waited longest since their last call
Round robin	each free agent in turn
Ring all	every free agent at once

- **Wrap-up** gives an agent time after a call before the next.
- **Longest wait**: callers who wait longer go to the queue's timeout destination. 0 waits for ever.
- Callers hear music, and their place in line.

The **Queues** screen is a live wallboard: who is waiting, for how long, and each agent's state.

Agents

Agents dial `*45` to log in or out of every queue, and `*46` to pause (a break) and resume; paused and logged-out agents are not offered calls. The wallboard has the same buttons, and shows why an agent paused.

State	Meaning
available	free; will be offered the next caller
ringing	being offered a call
busy	on a call
wrap-up	just finished; the wrap-up time is running
paused	on a break, with the reason given
away	in a meeting or on time off, from their calendar
logged out	not taking queue calls at all

Supervisors

On Live calls (chapter 32), **Supervise** rings a supervisor's phone; when they answer they **listen** (heard by nobody), **whisper** (heard by the agent only) or **barge** (heard by both), and can switch between them during the call.

Sticky queues and VIPs

Switch on **Sticky** in a queue and a returning caller (one answered in the last 30 days, or as many as you set) is offered first to the agent who answered them last time, if that agent is free; if they do not answer, the queue goes on as usual. **VIP callers**, at the foot of Queues, go to the front of every queue they call, after any VIP already waiting, and are marked VIP on the board.

CHAPTER 13

Menus (auto attendant)

In the web interface: `/menus`

A menu (an auto attendant) answers, plays a prompt and sends the caller where they press: 1 for sales, 2 for support, 0 for reception.

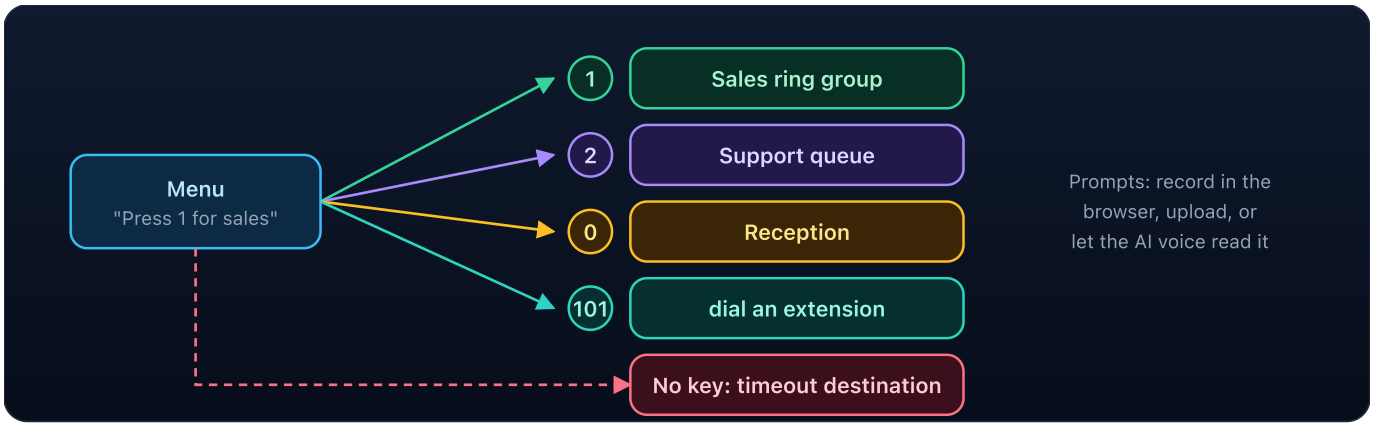


Figure 14. A menu (auto attendant): each key goes somewhere

Setting up a menu

1. On Menus (chapter 13) press **Add menu** and give it a name (and a number, if staff should be able to dial it).
2. Add **options**: a key, a label, and where the key goes.
3. Record the **prompt** in the browser, or upload a recording in any format (it is converted to telephone sound). Say every option: "For sales press 1, for support press 2."
4. Choose what happens when the caller presses nothing: the **timeout destination**, after the prompt has played the number of **repeats** you set.
5. Point an incoming route (or a schedule) at the menu.

Choices

- **Callers may dial an extension**: callers who know the number type it during the prompt.
- **Wait for a key**: seconds after the prompt before it repeats.
- A key can go to another menu, for a second level ("for billing questions press 1").

> Keep menus short: four options at most, the most common first, and always a way to a person. Callers who are lost hang up.

Prompts in other voices

With AI switched on (Settings), a prompt can be written as text and read by the AI voice instead of recorded.

CHAPTER 14

Schedules

In the web interface: `/schedules`

A schedule sends calls one way when you are open and another when you are closed. Point an incoming route at a schedule, and the schedule at the menu (open) and voicemail (closed).

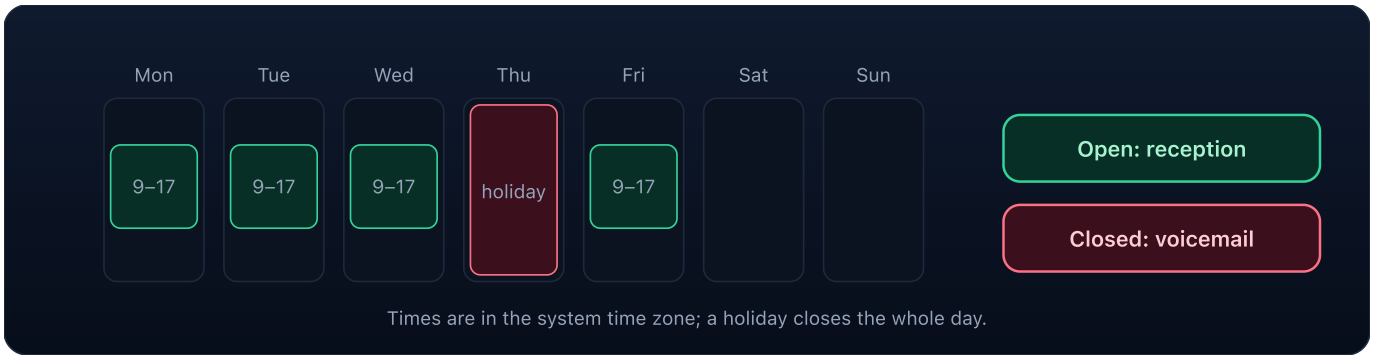


Figure 15. A schedule: open hours, holidays, and where calls go in each

Open hours

Each line of hours is a set of days and a time from and to. Several lines are allowed: Monday to Friday 9:00 to 17:00, and Saturday 10:00 to 13:00. Hours may cross midnight (22:00 to 06:00) for night shifts.

Holidays

Holidays count as closed all day, whatever the hours say. Add each date with a name; the name is shown on Call flow (chapter 10) when you try a call on that day.

Time zone

The time zone is the one in Settings. When the clocks change, schedules follow the local time.

> Try a call on Call flow (chapter 10) with a time and date (a holiday, a Saturday night) to see which side of the schedule it takes, before it matters.

CHAPTER 15

Scripts (JavaScript and Lua)

In the web interface: `/scripts`

A script decides where a call goes when menus, schedules and groups are not enough: callers from one area to one office, VIP numbers straight to a person, a decision from your own data. Scripts are written in JavaScript or Lua on Scripts (chapter 15).

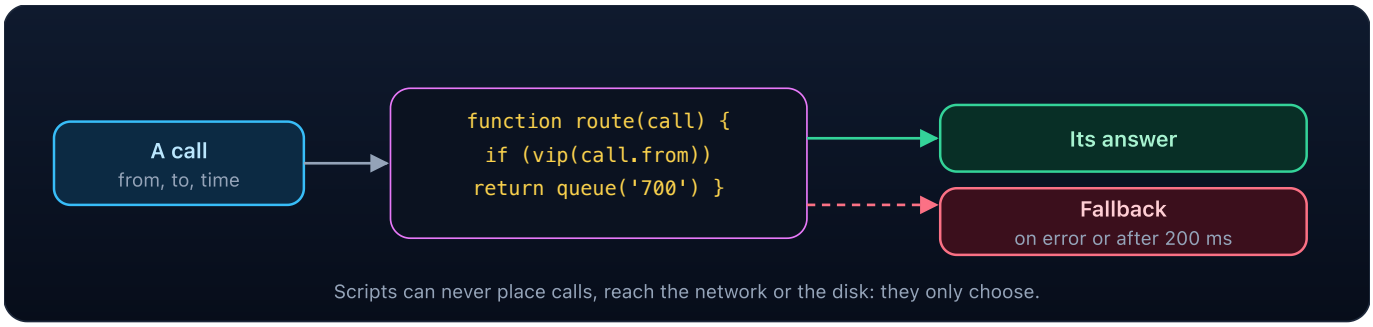


Figure 16. A routing script: code decides, a fallback is always there

Safe by design

- A script runs for at most 200 ms. If it fails or takes too long, the call goes to the script's **fallback**, so a mistake never loses a call.
- It runs in a sandbox: it cannot place calls, reach the network or the disk. It only chooses.
- Emergency calls never run scripts.

What a script sees

`call` has:

Field	What it is
`from`, `fromName`	the caller's number and name
`to`	the number dialled
`direction`	`inbound` or `internal`
`trunk`	the trunk it came in on
`weekday`, `hour`, `minute`	the time now (0 is Sunday)
`date`	YYYY-MM-DD

What it returns

A destination, `{type, target}`: type is `extension`, `ringgroup`, `queue`, `menu`, `schedule`, `voicemail` or `hangup`, and target is its number. Returning nothing uses the fallback. `log("text")` writes to the script's run list.

JavaScript

```
if (call.from.startsWith("212")) {
  log("New York caller")
  return { type: "ringgroup", target: "600" }
}
if (call.weekday === 0 || call.weekday === 6) return { type: "voicemail", target:
"100" }
```

Lua

```
if string.sub(call.from, 1, 3) == "212" then
  log("New York caller")
  return { type = "ringgroup", target = "600" }
end
if call.weekday == 0 or call.weekday == 6 then
  return { type = "voicemail", target = "100" }
end
```

Using a script

Point an incoming route, a menu key or a schedule at the script, as at any destination. **Test** runs it with a call you describe, without placing one; the run list shows each real run with its decision, time taken and any error. The call flow (chapter 10) runs scripts too when you try a call.

CHAPTER 16

AI receptionist, voicemail triage and asking

In the web interface: `/receptionists`

AI features use the AI service set in Settings (chapter 9): an OpenAI-compatible service (OpenRouter by default) or one you run yourself. What they need is sent there, so choose a service you trust with calls; for a 911 centre, its own government cloud counts as local.

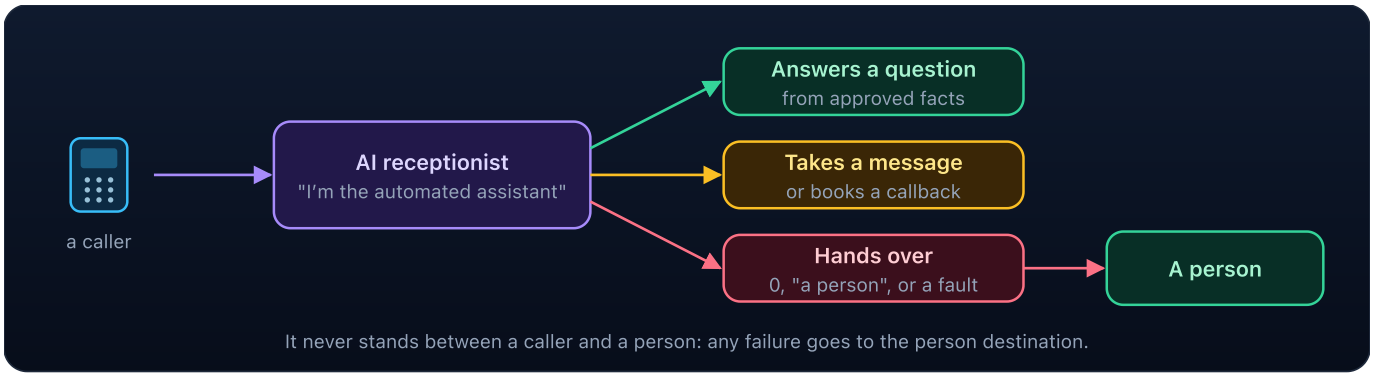


Figure 17. The AI receptionist: answers, helps, and always hands over to a person

AI receptionist

Choose a receptionist as where calls go: a route, a menu option or an extension's **If nobody answers**. It says it is automated, answers only from **Approved information**, takes a message, books a callback (see Callbacks (chapter 18)), or transfers to **A person is**. Pressing **0** or asking for a person always reaches someone, and so does any problem with the AI service or the speech model: callers are never stuck with it.

Models

Model does the text work: summaries, the receptionist's replies and asking in plain words. Turning speech into text needs a model that takes audio; set it as **Speech-to-text model** (for example google/gemini-2.5-flash on OpenRouter) to use a text-only model for the rest, such as meta-llama/llama-3.3-70b-instruct.

Voicemail triage

With **Transcribe and summarise voicemail** on, each message is also marked urgent, normal or low priority, with its kind (sales, support, billing...). Urgent messages raise an alert on every administrator's screen.

Ask in plain words

On Changes (chapter 46), describe a change ("send sales calls to voicemail after 5 pm"). The AI proposes it, BitVoice checks it like any other change and shows exactly what would change. Nothing changes until you press **Apply**, and it becomes a revision you can restore. Passwords are never sent to the AI.

CHAPTER 17

Voicemail

In the web interface: `/voicemail`

Each extension can have a mailbox. Callers who are not answered hear its greeting and leave a message; the person hears it on their phone, reads it on the screen, or gets it by email.

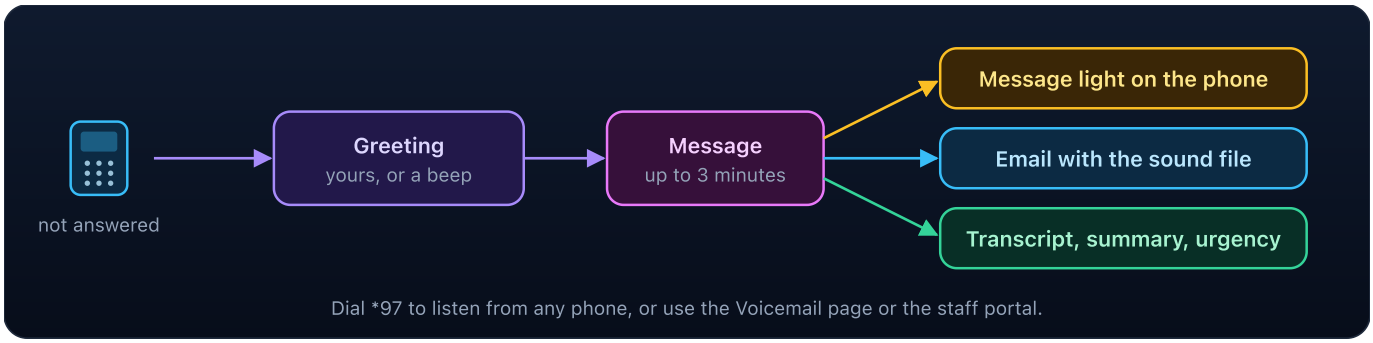


Figure 18. Voicemail: from a missed call to a message you can read

Switching it on

On the extension, switch on **Voicemail**. **If nobody answers** is then set to the mailbox. Give it a **PIN** (4 to 10 digits): it opens the mailbox from another phone, and signs the person in to the staff portal and at hot desks.

Setting	What it does
Longest message	10 to 900 seconds (180 by default)
Email new messages	to the extension's email address, with the message attached
Greeting	recorded in the browser, uploaded, or recorded from the phone; without one, callers hear a beep

Listening

- From the extension's own phone, dial `*97`: messages play newest first; keys replay, delete or keep each one.
- From any other phone, dial `*97`, enter the mailbox number and its PIN.
- On Voicemail (chapter 17) (administrators) or in the staff portal (the person): play, mark heard, delete.
- The phone's message light is on while there are new messages.

With AI

With **Transcribe and summarise voicemail** on (Settings, AI), each message is written out, summarised in a line, and marked urgent, normal or low priority with its kind (sales, support, billing). Urgent messages raise an alert on every administrator's screen. See AI (chapter 16).

Sound clean-up

Sound clean-up (Settings) removes steady background noise and evens out levels as messages are saved, so a message from a car or a busy street is easier to hear.

> Messages under a legal hold (chapter 57) cannot be deleted, by anyone, until the hold is released.

CHAPTER 18

Callbacks

In the web interface: `/callbacks`

People who asked to be rung back. An AI receptionist books one when a caller asks: it collects their name, the number to call (offering the one they are calling from), a good time and what it is about, reads them

back, and books it only when the caller agrees. It never promises an exact time. Administrators can also add callbacks by hand.

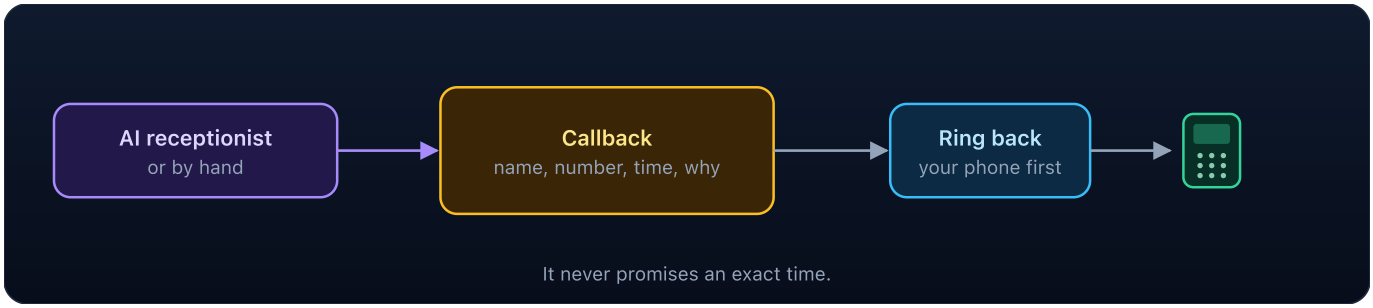


Figure 19. Callbacks: booked by the receptionist, rung back by a person

A new callback shows an alert on every open screen, and is emailed to the owner of the receptionist's **Take messages for** mailbox when that extension has an email address and email is set up in Settings.

To ring back, choose your extension and press **Ring back**: your phone rings first, and when you pick up BitVoice calls the number for you, with your normal routes and caller ID. Then add a note if you like and press **Done** (or **Cancel it**). Finished callbacks are kept below the open ones and can be opened again.

CHAPTER 19

Call links and website calling

In the web interface: `/call-links`

A call link is a web address that rings your business from any browser: no phone number, no app. Put it in an email signature, print its **QR code** on a flyer or a shop window, or paste the **website button** into your site.

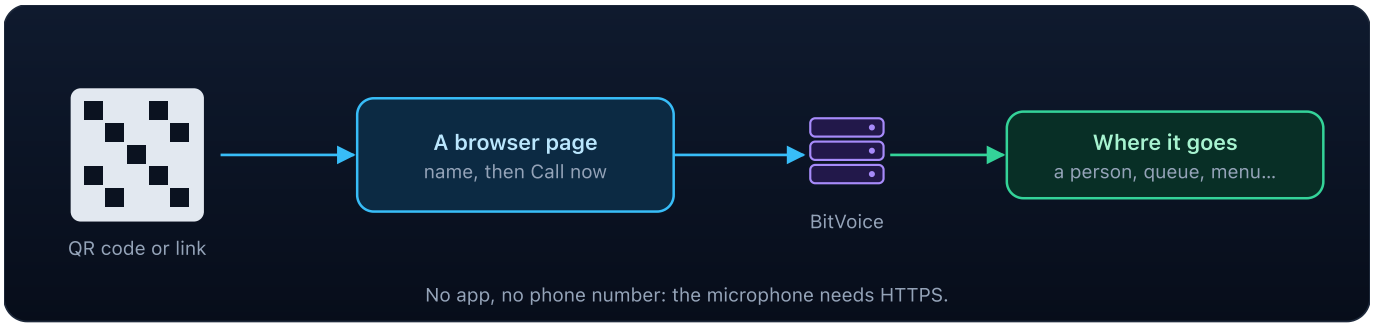


Figure 20. Call links: a web address that rings the business

On Call links (chapter 19), add a link, choose where its calls go (an extension, a group, a queue, a menu, the AI receptionist, a schedule or voicemail), and save: the page shows its address and QR code. The caller

sees the title and words you set, gives their name if you ask for it, allows the microphone and presses **Call now**; the keypad works for menus. Whoever answers sees **Web:** and the caller's name.

Switch **Open** off to close the line (the page says so). **Calls at once** limits how many people can call through one link (5 unless set). The address holds a long random part so it cannot be guessed; repeated wrong addresses are blocked like failed sign-ins. Callers' microphones need the page to be served over HTTPS, so give out the HTTPS address.

PART 3

Everyday calling

What people do on their phones every day, and what they set for themselves: forwarding, follow-me, pickup, conference rooms, paging, hot desks, faxes and texts.

CHAPTER 20

Feature codes, forwarding and speed dials

In the web interface: `/codes`

Feature codes are what people dial on their phones to change their own settings and reach features. They work on every phone, desk phone or softphone, with nothing set up on the phone itself. A rising tone means done; a falling tone means refused.

The codes at a glance

Code	What it does
`*72`	forward every call (then the number and #, or the number straight after: `*722125550199`)
`*73`	stop forwarding every call
`*90` / `*91`	forward when on a call / stop
`*52` / `*53`	forward when not answered / stop
`*78` / `*79`	do not disturb on / off
`*76`	switch do not disturb (rising tone: now on)
`*21`	switch follow-me on and off
`*8`	answer a call ringing in your pickup group
`**` and an extension	answer the call ringing that extension
`*80` and an extension	intercom: that phone answers on its speaker
`*0` and a code	a speed dial
`*43`	the echo test
`*97`	your voicemail
`70`	park a call (transfer it here); 71 to 79 pick it up
`*45` / `*46`	queue agents: log in or out / pause or resume
`*33` / `*34`	sign in / out at a hot desk
`*7` / `*8` during a call	pause and resume the recording (card payments)

Forwarding

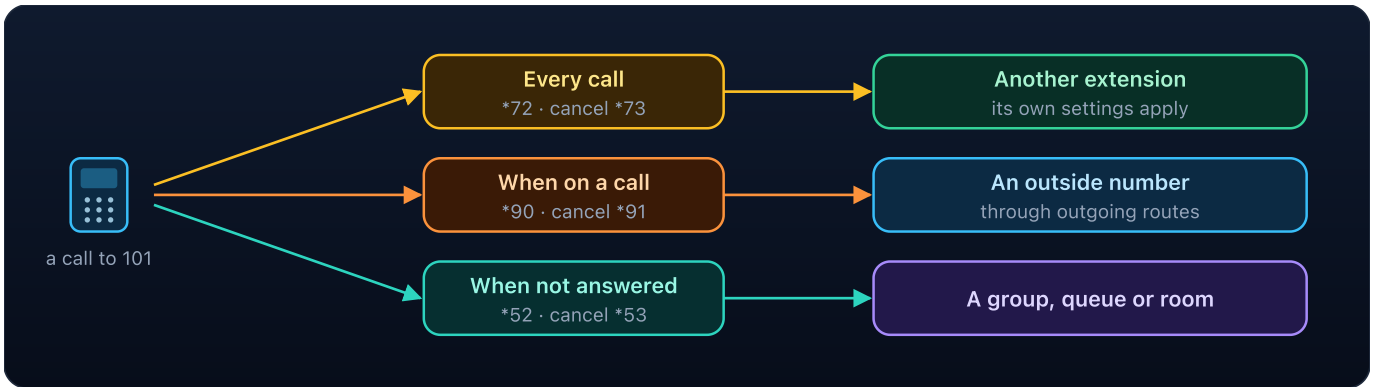


Figure 21. Forwarding: every call, when busy, when not answered

- `*72` forwards every call. Dial it, hear a tone, then dial the number and #. Or dial the number straight after the code.
- `*90` forwards calls that come while you are on a call. `*52` forwards calls you do not answer within your ring time.
- `*73`, `*91` and `*53` cancel them.

Forwarding to another extension uses that extension's own settings (its voicemail, its follow-me), up to four hops so two people forwarding to each other cannot loop. Forwarding to an outside number goes through the outgoing routes, as if the person dialled it, and needs the extension to be allowed outside calls.

Forwarding is live state, not configuration: setting it makes no revision, and it is kept across restarts. Extensions (chapter 2) and the staff portal show and change it too.

Do not disturb

`*78` on, `*79` off, `*76` switches. With do not disturb on, calls go to voicemail, or to the extension's no-answer destination. Do not disturb set by hand wins over the calendar (Calendars (chapter 26)). Emergency call-backs still ring.

Follow-me

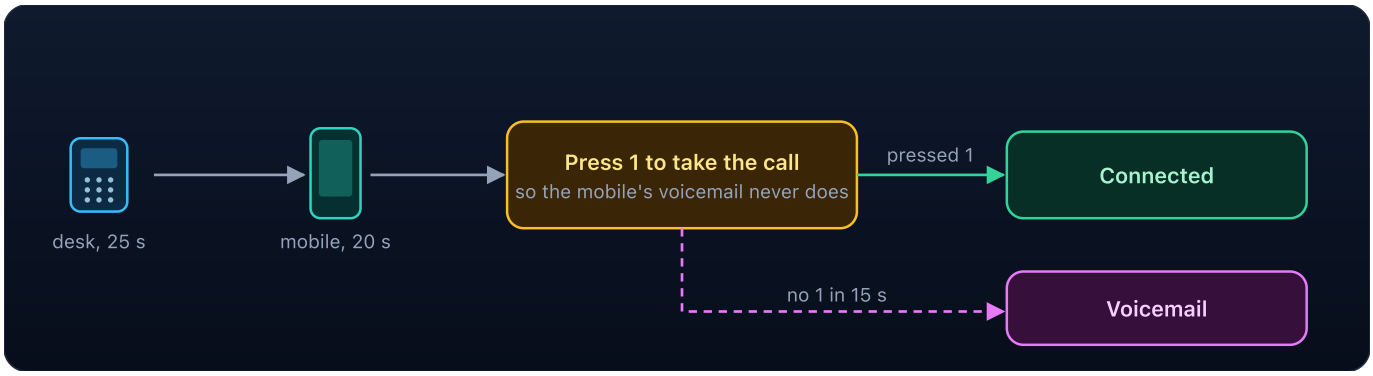


Figure 22. Follow-me: the desk first, then the mobile, then voicemail

`*21` switches follow-me on and off, for extensions that have follow-me numbers set (on the extension, or in the staff portal). The desk rings first for its ring time, then the follow-me numbers, one after another or all together. On an outside number with **press 1** on, the person hears a tone and presses 1 to take the call; a voicemail or a silence never does, and after 15 seconds the call moves on.

Answering someone else's call



Figure 23. Call pickup: answer a colleague's ringing phone

- `*8` answers the oldest call ringing anyone in your **pickup group** (set on each extension; names are compared without regard to capitals).
- `**` and an extension answers a call ringing that extension: `**101`.

Intercom

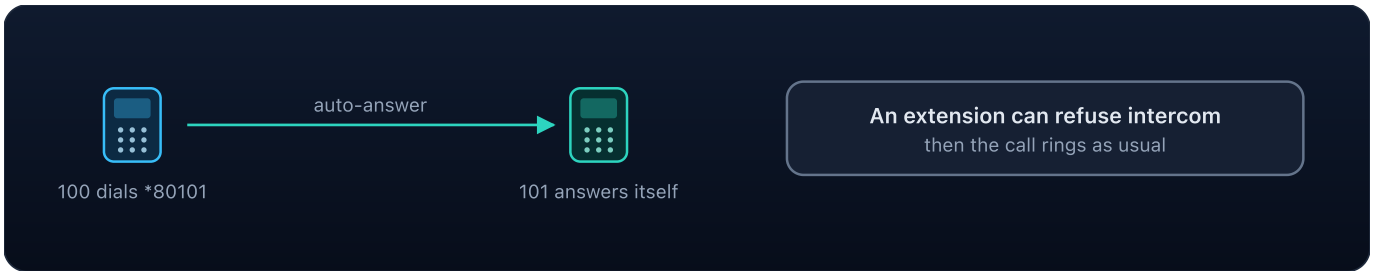


Figure 24. Intercom: the called phone answers on its speaker

`*80` and an extension calls it with auto-answer: its phone answers on the speaker, and both sides talk. Yealink, Grandstream and Snom read BitVoice's Call-Info header, Poly reads Alert-Info; some phones need intercom allowed in their own settings. An extension can refuse intercom; then the call rings as usual. Door stations never take intercom.

Speed dials

`*0` and a code dials a saved number: `*05`. The company's speed dials are on this page; each person can have their own on their extension or in the staff portal, and theirs win over the company's with the same code. Codes are 1 to 3 digits.

Changing the codes

Every code can be changed on Feature codes (chapter 20), or switched off with the word `off`. Two features cannot share a code, and no code may be an extension's or a group's number: BitVoice says which one clashes. The echo, voicemail and park numbers are in Settings (chapter 9).

CHAPTER 21

The staff portal

The staff portal lets everyone look after their own calls without an administrator: do not disturb, forwarding, follow-me, speed dials, voicemail and its greeting, their PIN, and a list of their own calls. It is at `#/me` on this server's address, and linked from the sign-in page.

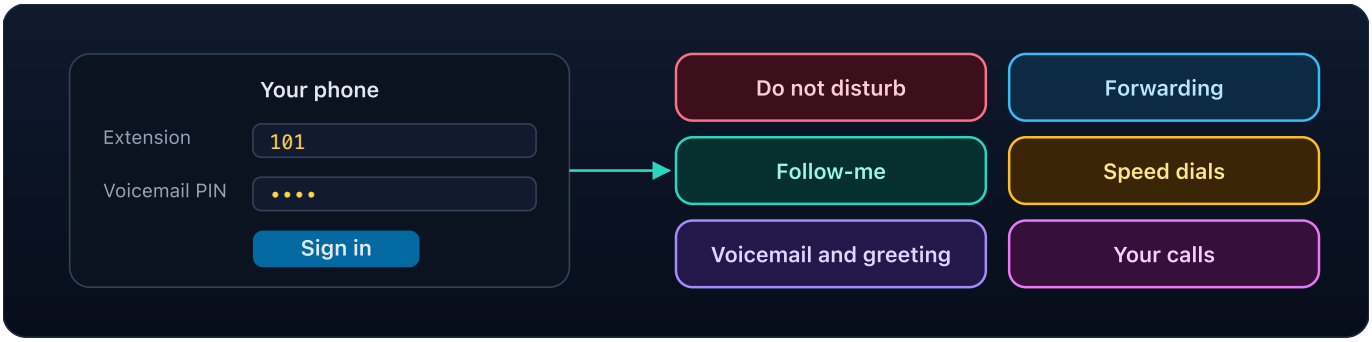


Figure 25. The staff portal: everyone looks after their own calls

Signing in

People sign in with their **extension number** and their **voicemail PIN**: the same PIN that opens their voicemail from a phone and signs them in at hot desks. An extension without a PIN cannot sign in; an administrator sets one on the extension.

Repeated wrong tries from one address lock it out for ten minutes. A portal session reaches only the person's own settings: never another extension, the configuration or anyone's recordings.

What people can do

Tab	What is there
Where calls go	do not disturb; forwarding of every call, when busy and when not answered; follow-me numbers; own speed dials
Voicemail	messages to play and delete, with transcripts and summaries when AI is on; record or upload the greeting
Your calls	the person's own calls, newest first, without recordings
PIN	change the PIN, giving the current one

Forwarding and do not disturb set here are the same as those set from the phone with feature codes (Feature codes (chapter 20)): whichever was set last is in force. Follow-me numbers and speed dials are part of the extension's configuration, so changing them makes a revision on Changes (chapter 46), recorded as the person's own.

> Forwarding to an outside number works only for extensions allowed to call outside numbers. The portal says so rather than saving a forward that could never ring.

CHAPTER 22

Conference rooms

In the web interface: `/conferences`

A conference room is a number people dial to all talk together. Staff dial it from their desks; outside callers reach it through an incoming route or a menu option that points at the room.



Figure 26. A conference room: PIN, leaders, and who is talking

Setting up a room

1. Open Conference rooms (chapter 22) and press **Add conference room**.
2. Give it a **number** (any free number, 500 by default) and a **name**.
3. Optionally set a **PIN**. Callers hear a tone, enter the PIN and press #. Three wrong PINs and the call ends.
4. Optionally set a **leader PIN**, different from the PIN. Whoever enters it joins as a leader.
5. To let outside callers in, point an incoming route (or a menu key) at the room.

Choices

Choice	What it does
Wait for the leader	everyone else hears the hold tone until a leader joins; when the last leader leaves, the room closes and everyone is hung up. Needs a leader PIN.
Members join muted	for briefings and lectures: leaders talk, members listen. A member presses 1 to ask to speak, and a leader unmutes them.
Record the room	the whole room is kept as one recording in Call history (chapter 33), unless recording is off for the whole system
Tones when people join and leave	on by default
Most people at once	50 by default, up to 200

Keys during a call

- `1` mutes and unmutes yourself (in a room where members are muted, it asks to speak).
- `2`, for a leader, locks the room so nobody new can join, and unlocks it.

While a room is in use

The page shows each room in use: who is in, who is talking (the green light), who is muted, who has asked to speak, and who is still waiting for the leader. An administrator can mute or unmute anyone, remove someone, and lock or unlock the room.

Everyone in a room hears everyone else who is not muted, mixed by BitVoice; each person's phone can use its own codec. A call record is kept for each person's call, and one for the room itself with its length and recording.

> Conference sound is mixed at telephone quality (8 kHz), whatever codecs the phones use: it is what every phone can hear, and it keeps a room of fifty cheap to run.

CHAPTER 23

Paging and overhead speakers

In the web interface: `/paging`

Paging speaks through many phones at once: dial a paging number, and every phone in the group answers on its speaker by itself. Use it for announcements to a warehouse, a school, a shop floor.

Setting up a paging group

1. On Paging (chapter 23) press **Add paging group** and give it a number (800 by default) and a name.
2. Choose the phones. Overhead speakers set up as extensions (kind **Overhead speaker**) can be members too.
3. Optionally, a **multicast** address and port.

Multicast

Multicast also sends the page to a multicast address and port (`239.0.1.10:5000`), as G.711 µ-law, which overhead speakers and many desk phones can listen to without a call each. A group may have only a multicast address. It works on one network: routers do not pass multicast on unless they are set up to.

Paging and intercom

	Paging	Intercom
Who hears	everyone in the group	one extension
Talk back	no, one way	yes, both ways
How	dial the group's number	`*80` and the extension

Phones that do not answer a page by themselves within a few seconds are left out, so one phone without auto-answer never holds a page up. Some phones need auto-answer (intercom) allowed in their own settings.

CHAPTER 24

Hot desking

In the web interface: `/phones`

A hot desk is a shared phone that becomes whoever signs in at it: for offices where people do not have fixed desks. Make one by switching on **Hot desk** in the extension the phone is set up as (the desk's own extension, such as 900 Desk 3).

Signing in and out

1. At the phone, dial `*33` and wait for the tone.
2. Enter your extension and #, and wait for the tone.
3. Enter your voicemail PIN and #. A rising tone means you are signed in; a falling tone means it was refused.

From then on, calls to your extension ring this desk as well as your other phones, and calls you make from it show your number and name and count as yours on Costs. Signing in at another desk moves you there. `*34` signs you out. Five wrong tries lock the desk for ten minutes.

> Emergency calls from a hot desk always carry the desk's own site, floor and room, not your usual ones: responders go to where the phone is.

Managing hot desks

On Phones (chapter 3), **Hot desks** shows who is signed in where; administrators can sign someone in (without their PIN) or out. The codes, and a time each day when everyone is signed out, are in Settings. Sign-ins are kept across restarts.

CHAPTER 25

Contacts on every phone

In the web interface: `/contacts`

Contacts is the company directory: the suppliers, clients and other offices people call. Every phone lists it beside the extensions, and when one of these numbers rings in, phones and the caller card show the contact's name (when the provider sent none). A number matches however it is written: +1 212 555 0100 and 2125550100 are the same line.

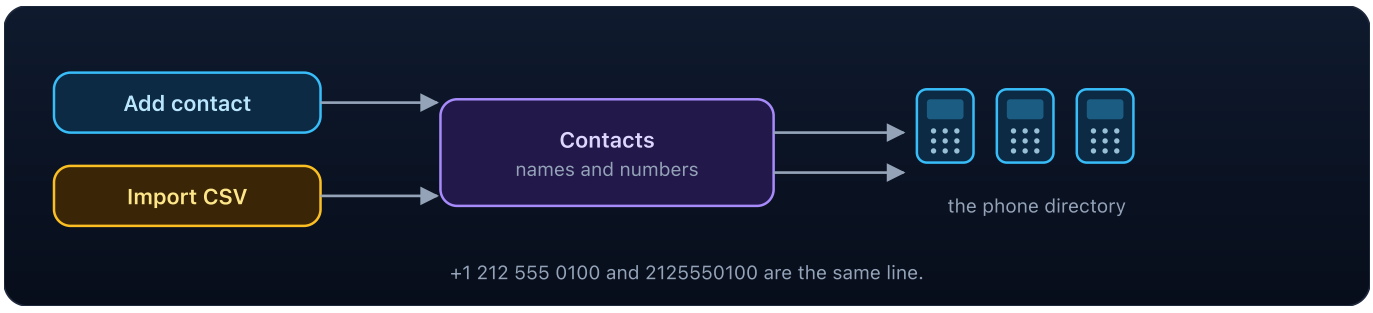


Figure 27. Contacts: one directory on every phone

Adding contacts

Add contact for one at a time, with as many numbers (work, mobile, home, other) as they have. **Import** takes a spreadsheet saved as CSV whose first row names the columns: name, company, work (or phone), mobile, home, other, email, notes. BitVoice checks the whole file first and says which lines are wrong; a contact with the same name as one there is updated rather than added twice, and **Replace every contact** starts the list afresh. **Export** downloads the list as CSV.

On desk phones

Phones set up by Phones (chapter 3) (provisioning) get the directory with no more setup:

- **Yealink**: a remote phonebook named Company, refreshed every hour.
- **Grandstream**: the phonebook download, every hour.
- **Polycom**: the directory file, read when the phone starts.
- **Snom**: the list is written into the phone's settings, so it updates when the phone next checks them.

In the browser phone

Contacts under the keypad finds a name or number and puts it ready to call.

CHAPTER 26

Calendars and do not disturb

In the web interface: `/extensions`

An extension can follow its person's calendar: during meetings and time off, its calls go to voicemail, or are forwarded (to another extension, or a mobile), instead of ringing a phone nobody will answer. Ring groups ring the others meanwhile, and in queues the agent shows as **Away**.

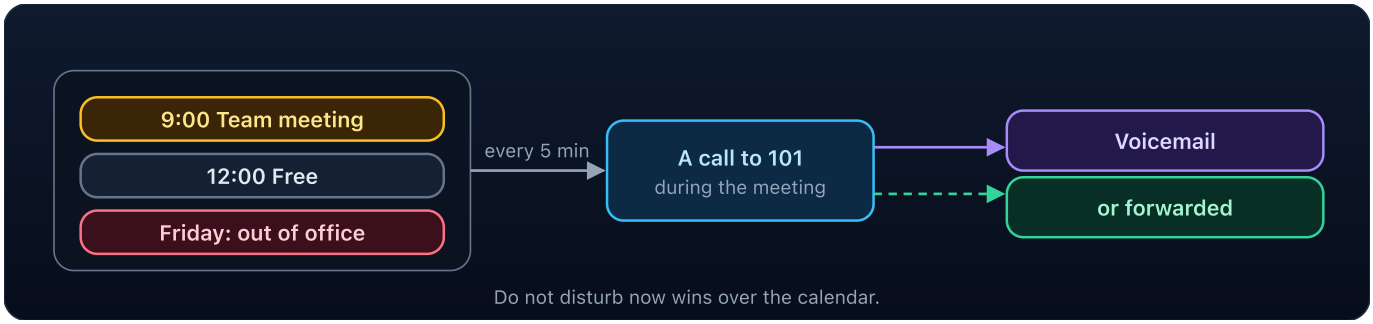


Figure 28. Following a calendar: no ringing in meetings

In the extension, **Calendar and do not disturb**, paste the calendar's private address:

- **Outlook**: Settings, Calendar, Shared calendars, Publish a calendar; choose the calendar and "Can view all details", then copy the ICS link.
- **Google Calendar**: Settings, the calendar on the left, then **Secret address in iCal format**.

Check reads it at once and lists the next busy times. BitVoice reads each calendar every five minutes. Time off (all-day events, and Outlook's "out of office") always counts; meetings count when **Meetings count as busy** is on; events marked free never count, nor do cancelled ones. Recurring meetings, with skipped and moved occurrences, are understood. If a calendar cannot be read, Health says so.

Do not disturb now stops the extension ringing until switched off, whatever the calendar says. The extensions list shows who is away and until when.

The private address works like a password: anyone with it can read that calendar, so it is kept with the other secrets.

CHAPTER 27

Mobile apps

The BitVoice apps for iPhone and Android make a mobile phone one of an extension's phones. It rings like a desk phone, on the phone's own call screen, even when the app is closed.

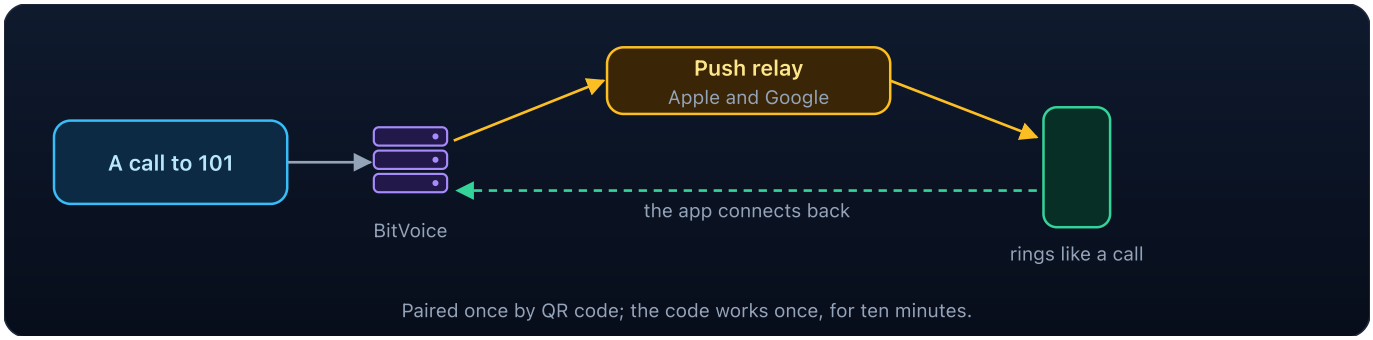


Figure 29. The mobile apps: woken by a push, rung like a desk phone

Setting up push

A closed app is woken for calls by a push message. In Settings (chapter 9), **Mobile apps**, choose one:

Push	What you need
Push relay	BioDream's relay (with hosting or support) or your own `bvpush`: its **Relay address**, **Key ID** and **Secret**
Own apps	your own builds: for Apple, the **Team ID**, **Bundle ID**, **Key ID** and **Key (.p8)**; for Google, the **Firebase service account (JSON)**

Set **Address the apps use** to this server's https address as it is reached from anywhere, not only the office.

> With push off, the app rings only while it is open.

Pairing a phone

1. Open the extension and press **Add a phone**.
2. Scan the QR code in the app. The code works once, for ten minutes.

When a call comes in

BitVoice wakes the app; it connects, rings on the phone's own call screen, and is answered like any other phone of the extension: ring groups and queues ring it with the extension's other phones. The extension's page lists its paired phones, whether each **Can be woken for calls**, when it was last seen and last woken, and any push problem. **Unpair** stops a phone ringing at once, for a lost phone.

CHAPTER 28

Video calls

In the web interface: `/settings`

When both phones in a call send video (video desk phones, softphones, video doorbells), the picture passes between them through BitVoice. Nothing to set up: it is on unless **Settings → Video calls** is switched off.

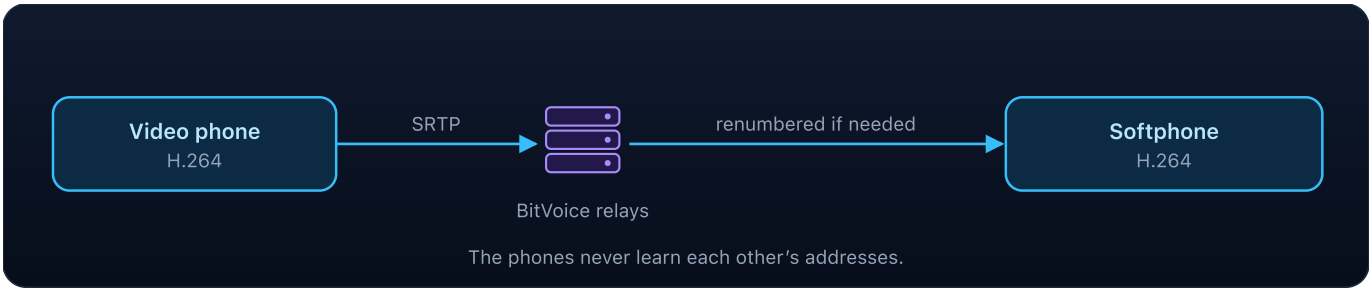


Figure 30. Video calls: relayed as they are

- BitVoice relays the video as it is, on a port of its own for each phone, so the phones never learn each other's addresses. It never converts video: the two need a codec in common (nearly every video phone has H.264). When they number the same codec differently, BitVoice renumbers it on the way.
- The picture is encrypted (SRTP) to each phone that encrypts its sound.
- A phone's request for a fresh picture (after lost packets) reaches the other phone, so the picture recovers in a moment.
- A camera switched on during a call (a softphone's video button) works too: the other phone is asked first, and the picture starts when it says yes.
- A phone without video, or a call to an outside number, gets sound only. On hold or while parked the picture pauses, and comes back when the call does. After a transfer it carries on only when the new party sends video too.
- Video uses far more of the network than sound (about 0.5 to 2 Mbit/s each way per call): allow for it on slow links and in the media port range.

CHAPTER 29

Fax: sending and receiving, fax machines and analogue phones

In the web interface: `/faxes`

Fax machines, analogue phones, door phones and old lines connect through a SIP gateway: an ATA (Grandstream HT series, for one or two analogue phones or a fax), or a bigger gateway (Patton SmartNode, AudioCodes MediaPack and Mediant) for many analogue ports, an ISDN line (BRI/PRI) or old analogue lines (FXO). Each port registers to BitVoice as an extension, like a phone.

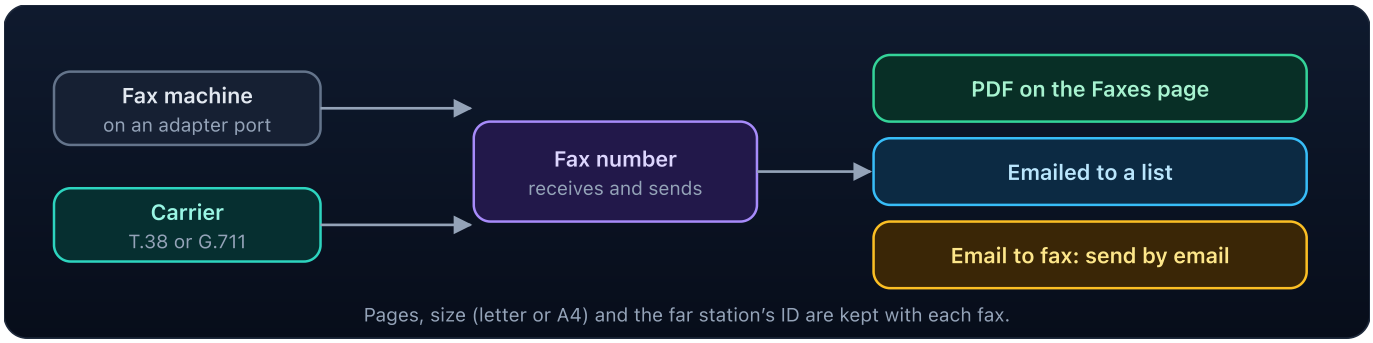


Figure 31. Fax: machines, email and PDFs

Receiving faxes without a fax machine. On **Faxes**, add a fax number (300, say), with the email addresses each fax goes to. Then in Call routing (chapter 5) send a phone number's calls to it. BitVoice answers as a fax machine, switches the call to T.38 and takes the fax; each fax becomes a PDF, kept on Faxes (**View**, **PDF**) and emailed. A fax that breaks off part way keeps the pages that arrived and says so. This needs a provider that supports T.38 (most do); a fax sent as sound cannot be received this way.

Sending a fax. On **Faxes**, **Send a fax**: choose the fax number it goes from, type the number to send to, write a note if you like (it becomes the first page) and add documents. Text files and pictures (PNG, JPEG, GIF) work everywhere. PDF needs poppler on the server (`apt install poppler-utils`, `apk add poppler-utils`, `brew install poppler`); Word, Excel, OpenDocument and RTF need LibreOffice as well. The page says which the server can do. Every page gets a header line, as fax machines print it: the time, the fax number's name, its identity and the page count. Pages are US letter in North America and A4 elsewhere, from the country in Settings.

BitVoice calls the number over your outbound routes, as an extension would, and sends at the fastest rate the line carries (14,400 bit/s, slower if the line is poor). Faxes from one fax number go one at a time. A busy number, or one that does not answer, is tried again five minutes later, three times in all. Sent faxes are listed with the received ones: **Sent**, **Waiting to send**, **Failed** (with **Try again**), or **Part sent** when the line broke after some pages.

Email to fax. Below the list, set up a mailbox for BitVoice to read (IMAP over TLS, port 993; for Google and Microsoft mailboxes use an app password), then **Test** and **Save**. Anyone allowed to send writes to that mailbox with the fax number in the subject (`Fax 212 555 0199`), or to a plus address (`fax+2125550199@example.com`) if the mail server takes them. Attachments become the pages and the email's own words the cover note (the signature and quoted mail are left out). It goes from the fax number that lists the sender: its **Email each fax to** addresses may always send, and **May also send faxes by email** adds others (whole addresses, or `@example.com` for a domain). The sender gets an email when the fax has gone through, or why it could not.

Email is easy to forge, so BitVoice faxes a message only when your mail server found it really came from the sender's domain (DKIM, SPF or DMARC, in the Authentication-Results line it adds). Other messages

are dropped without a reply. Switch **Accept mail the sender's domain did not vouch for** on only for a mailbox no outside mail can reach.

Fax machines on gateways. BitVoice carries fax over T.38, the fax standard for the internet: when the gateway hears a fax tone it switches the call to T.38, BitVoice asks the other side (your provider, or another gateway) to switch too, and passes the fax between them. If the other side cannot do T.38, the call stays on G.711 and the fax goes as sound, which works on a good network. Live calls show **fax (T.38)** while a fax is going through.

For reliable faxes:

- Use T.38 on the gateway and choose a provider that supports it.
- Make the port's extension an **Analog adapter port** (Extensions, Kind): BitVoice then offers it only G.711, which faxes sent as sound need.
- On the gateway's fax port: Fax Mode T.38, only PCMU or PCMA, line echo canceller off, call waiting off, and a fixed, high jitter buffer.

CHAPTER 30

Business texting (SMS)

In the web interface: `/texts`

Send and receive text messages from your business numbers on **Texts**. Texts go through a texting provider. Built in: TextMagic, Twilio, SignalWire, Telnyx, Bandwidth, Vonage, Plivo, Flowroute and VoIP.ms, plus **Your own webhook** for any other service or a gateway of your own. Any number of provider accounts can be used at once: each number says which one carries it.

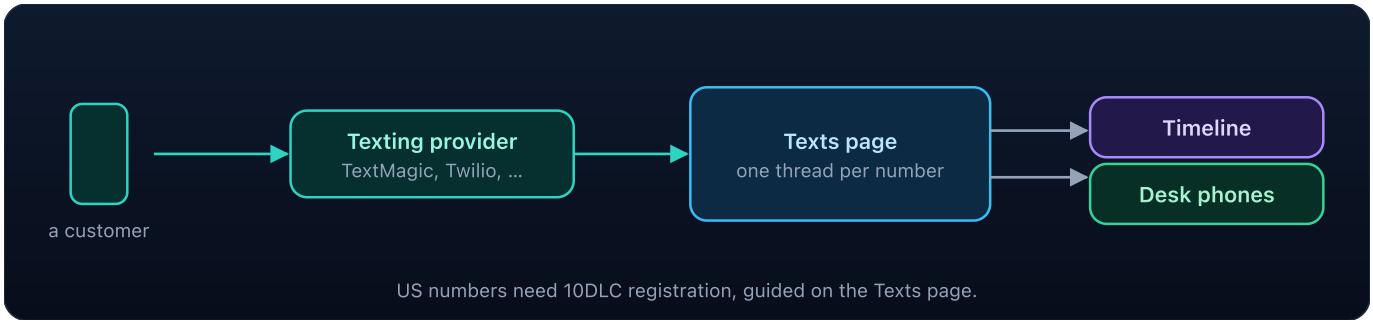


Figure 32. Business texting: the same numbers, one conversation view

Setting up

1. In the provider's portal, find the account's API details. The form asks for exactly what that provider calls them (Twilio: Account SID and auth token; Telnyx: an API key; Bandwidth: API user, password,

account ID and messaging application ID; SignalWire: also your space address) and says where to find them.
2. On Texts, **Texting setup**, add the provider with them, and save. BitVoice shows an address for received texts: set it in the provider as where received messages are sent (its callback, webhook or message URL). For VoIP.ms, add `?from={FROM}&to={TO}&message={MESSAGE}&id={ID}` to the end. Delivery receipts sent to the same address are ignored. BitVoice must be reachable from the internet there.
3. Add each number you text from (with its country code), the provider that carries it, and the extensions to tell by email when a text comes in.

US 10DLC

US carriers block texts from ordinary business numbers that are not registered. Register your brand and then a campaign with your texting provider; the checklist on Texts tracks where they stand, and warns when your sample message lacks the opt-out words carriers expect.

Conversations

Each conversation shows who it is (from Contacts), unread texts, and the thread. Enter sends; the counter shows how many texts a long message takes. When someone replies **STOP**, nothing more can be sent to them from that number until they reply **START**. Received texts also appear as a notice in BitVoice.

CHAPTER 31

Transfer briefings

In the web interface: `/settings`

So a caller never has to repeat themselves: with **Brief whoever a call is transferred to** on (Settings, AI), when a call is transferred, BitVoice writes a two-sentence summary of the call so far (who is calling, what they want, anything already promised) for the person it goes to. It appears on their browser phone with the ringing call, on Live calls (chapter 32), and as a text on their desk phone's screen (for phones that show SIP messages).

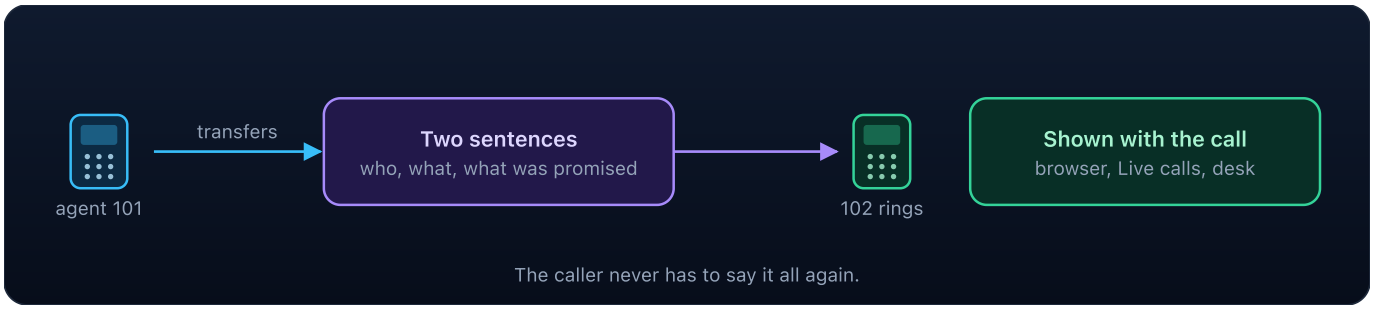


Figure 33. Transfer briefings: the next person knows already

It works the same when the AI receptionist hands a caller to a person: the briefing comes from what the caller told the assistant.

It needs the AI service and a speech-to-text model (Settings, AI), and the call's audio so far (up to five minutes) is sent to that service to be written out. A summary takes a few seconds, so it may appear just after the phone starts ringing.

PART 4

Customers and insight

Knowing who called and how it went: the live calls and history, the customer timeline, CRMs, recordings and consent, quality, costs and caller ID reputation.

CHAPTER 32

Live calls

In the web interface: `/calls`

Live calls shows every call in progress, updated as it happens: who called whom, how long it has been going, its codecs and its quality.

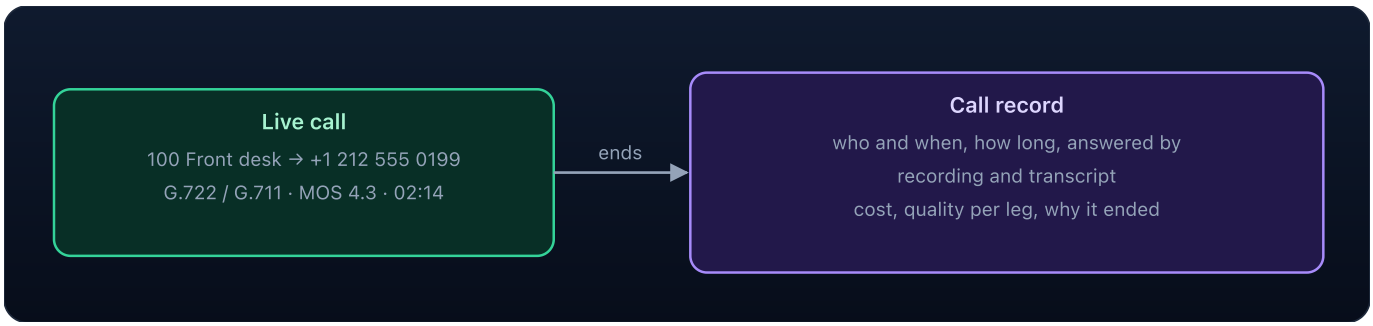


Figure 34. Live calls and history: every call, while it happens and after

What administrators can do

Action	What it does
Hang up	ends the call for both sides
Open the door	for a call from a door station
Pause recording	for a card number: the paused stretch is silent in the recording and marked in the history (Call recording (chapter 37))
Listen	hear the call silently
Whisper	talk to the agent only; the caller does not hear
Barge	join the call; everyone hears you

To listen, whisper or barge, press **Supervise**, choose the phone to use under **From phone** and press **Ring me**: your phone rings, and when you answer you join the call. You can switch between listening, whispering and barging while you are in it.

What else shows

- A caller card while an outside call rings (from your CRM, or past calls; see CRM (chapter 35)).
- The briefing written for the person a call was transferred to (Transfer briefings (chapter 31)).
- The caller ID verdict: **Verified**, **Partly verified** or **Unverified** (Spam protection (chapter 42)).
- With several servers, calls on the others too, read only: hang up or listen on the server that has the call.

When a call ends, it moves to Call history (chapter 33).

CHAPTER 33

Call history

In the web interface: `/history`

Every finished call: who called whom, when, for how long and how it ended. Sort the columns, play recordings (a pause for card numbers is marked), and download the list as CSV. How long records are kept is set in Settings, Call recording; a legal hold (chapter 57) keeps them longer.

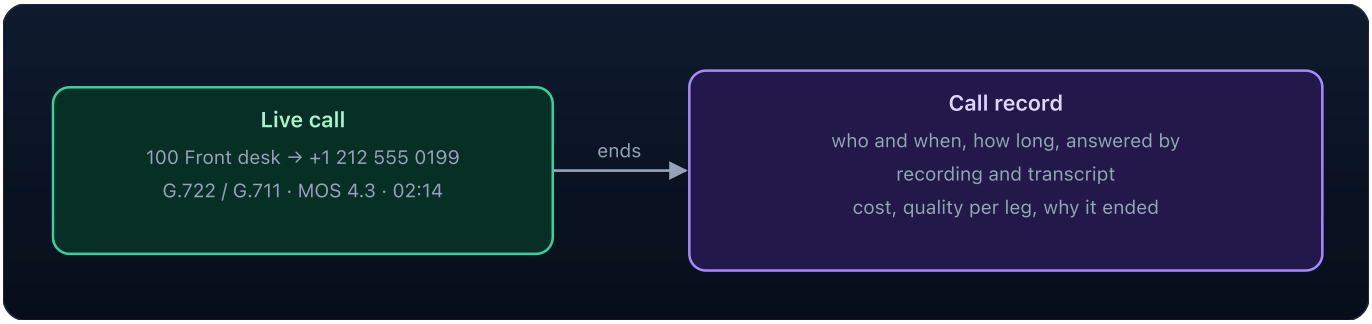


Figure 35. Live calls and history: every call, while it happens and after

Transcripts and search

With AI on and **Transcribe recorded calls** switched on (Settings, AI), every recorded call is written out, with a short summary and its action items: what someone agreed or was asked to do. Each side of the call is written out separately from the recording, so the transcript shows who said what, and when. The transcript button next to a recording opens it; administrators can also transcribe a single call, or transcribe one again.

Search what was said finds calls by words from their transcripts, summaries and action items. Only recorded calls are transcribed. A transcript is deleted with its recording, and reading or searching transcripts is noted in the audit log, like playing a recording.

CHAPTER 34

Customer timeline

In the web interface: `/timeline`

The customer timeline shows everything with one phone number over the last year, newest first, so whoever picks up knows the whole story.

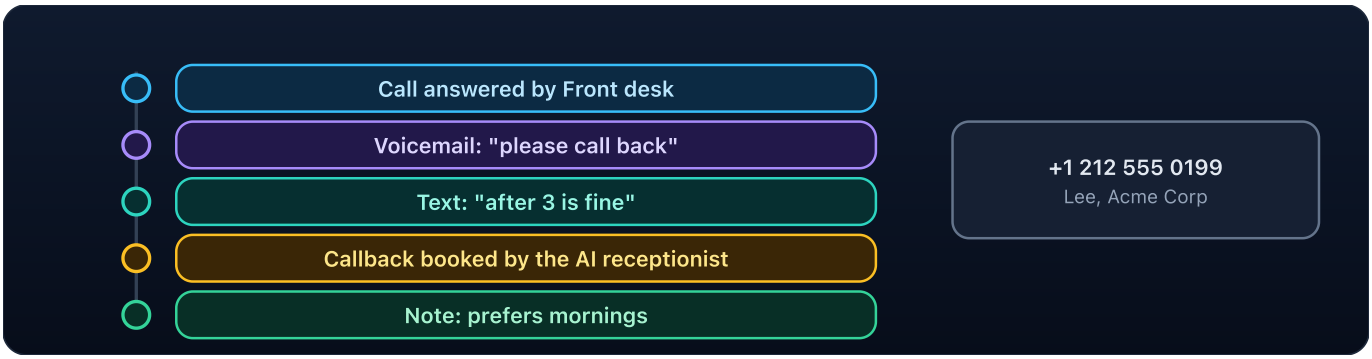


Figure 36. One customer timeline: everything about a number in one place

What it shows

Item	With
Calls	who took them, how long, and with a transcript, its summary, the recording and the transcript
Voicemails	their summaries
Texts	both ways
Callbacks	asked for, and whether done
Notes	the team's own notes

A number matches however it is written: +1 212 555 0100 and 2125550100 are the same.

Opening it

From a caller card (**Timeline**), from Contacts, from an incoming call in Call history, from a text conversation, or type a number at the top of Timeline (chapter 34).

Notes

Administrators can **Add a note for the team**: it shows on the timeline for everyone, with who wrote it and when, and can be removed.

CHAPTER 35

CRM: caller cards and call logging

In the web interface: `/settings`

Connect a customer system in Settings (chapter 9), CRM. It works with any system that has a web address, without code written for each product.

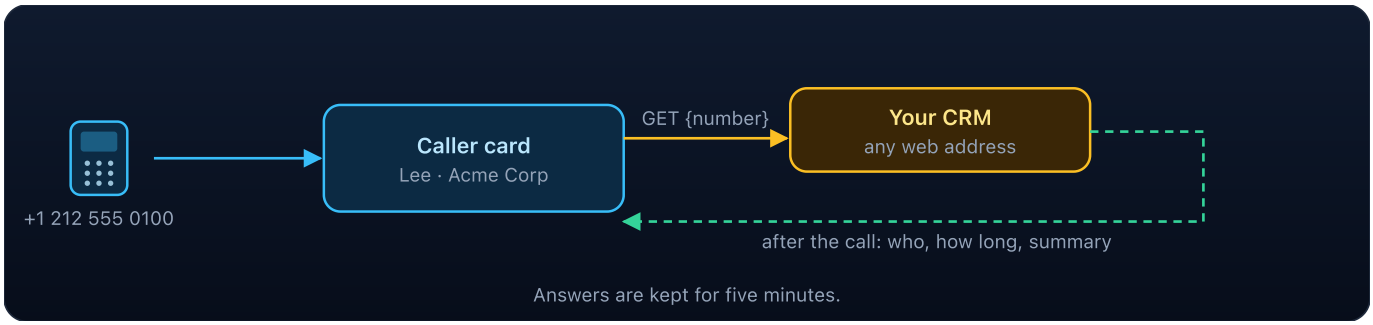


Figure 37. CRM: a card while it rings, a log after it ends

Caller cards

While a call from outside rings, Live calls shows a card: who it is (from your CRM, or the extension's name), how often they have called in 90 days, any callback (chapter 18) they are waiting for, and the summary of their last transcribed call. The browser phone shows a shorter card to the person answering: who it is, past calls and waiting callbacks, but not other people's call summaries.

For the name, set **Lookup address**: BitVoice asks it (GET) with the caller's number in place of {number}, sending **Lookup header** with its value (a vault reference works), and expects JSON with name, company, url and notes; 404 means unknown. Answers are kept for five minutes. **Open in CRM link** puts a link to the caller on the card. Use **Look it up** to try a number before saving.

Call logging

Set **Call log address** and **Signing secret**, and every call is sent there as it ends (event call.ended), again with its summary, action items and transcript once transcribed (call.summarized), and each booked callback (callback.new). Each POST carries X-BitVoice-Event, X-BitVoice-Timestamp and X-BitVoice-Signature: sha256= followed by the HMAC-SHA256, with the secret, of the timestamp, a dot and the body. A failed delivery is tried four times over about a minute; if it still fails, Health says so. **Send a test event** checks the address and the secret.

CHAPTER 36

Call insight for supervisors

In the web interface: `/quality`

For contact centres: evidence about how calls went, for a supervisor to listen to, never an automatic score.

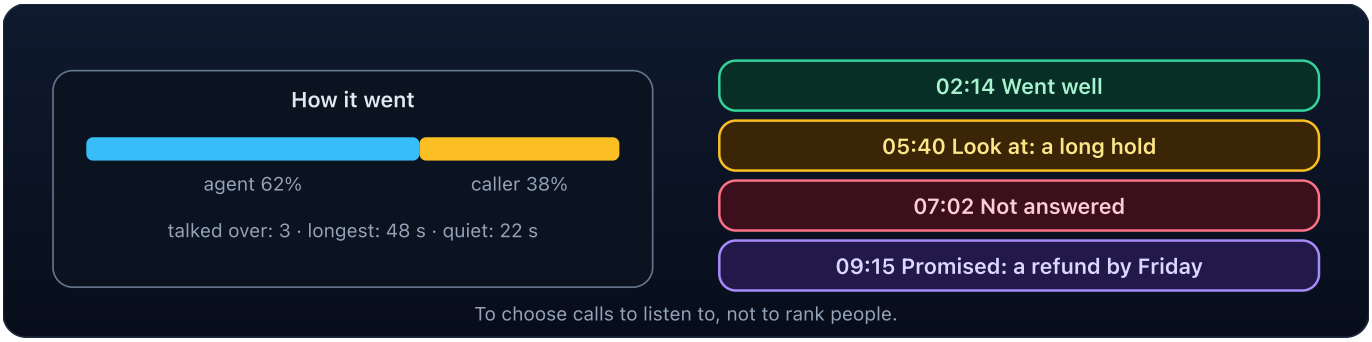


Figure 38. Call insight: evidence for a supervisor, never a score

In a call's **Transcript**, **How it went** shows, measured from the recording's sound alone: how much each side talked, each side's longest stretch, how often each started talking over the other, how long both spoke at once, and the quiet (a long silence is often a hold or dead air).

With **Note how each transcribed call went** on (Settings, AI), the AI also reads the transcript and says how the caller seemed at the start and the end, and lists up to six moments: **Went well**, **Look at** (talking over the caller, a long hold, an unclear answer), **Not answered** (a question the caller asked that got no answer) and **Promised** (something someone said they would do), each with the time and the words said, so you can go straight to it. An administrator can ask for it on any transcribed call with **Look at this call**.

On Call quality (chapter 36), **Conversations, by who took them** sums the transcribed calls of the period per person: their share of the talking, their longest stretch, how often they talked over the caller, quiet time, calls that ended with the caller upset, and the moments flagged. Use it to choose calls to listen to, not to rank people.

CHAPTER 37

Call recording and card payments

Recording is set in Settings (chapter 9), **Call recording**: off, every call, or only the calls of extensions marked **Record calls**. Recordings play from Call history (chapter 33), and with AI on they are written out and summarised.

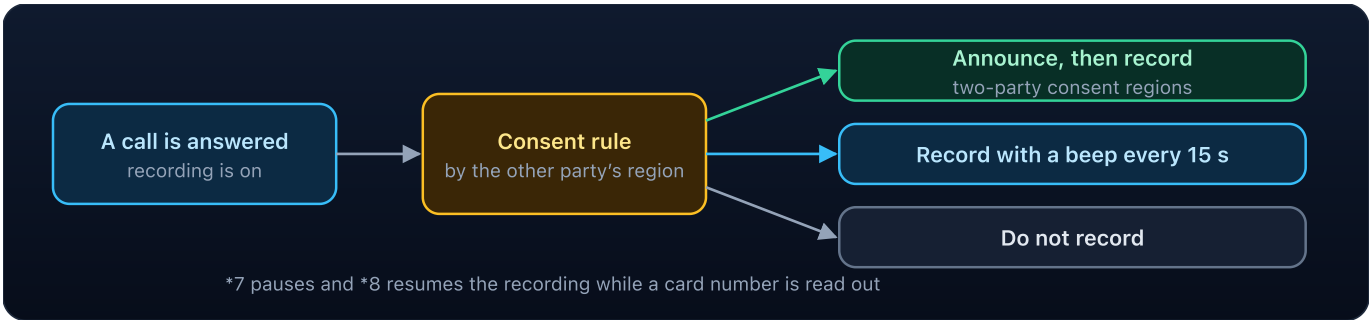


Figure 39. Recording with consent: announce, beep, or not at all

Telling people

Some places need everyone on a call to be told it is recorded: **Announce**, **Beep** and rules by region are in Recording consent (chapter 38).

Card payments (PCI DSS)

Card numbers must not be kept in recordings. Before a caller reads one out, the person on the extension presses `*7`; `*8` resumes. **Pause recording** on Live calls (chapter 32) does the same. The paused stretch is silent in the recording and listed in the call history, so an auditor can see where and for how long.

Conference rooms

A room with **Record the room** on is kept as one recording of the whole room (Conference rooms (chapter 22)).

How long things are kept

Settings sets how many days recordings, call history and voicemail are kept; older ones are deleted within the hour. Anything under a legal hold (chapter 57) is kept whatever its age. The archive (chapter 56) can keep copies elsewhere.

Administrators and viewers can play recordings and read transcripts; only administrators can transcribe a call, or transcribe it again. Playing a recording or reading its transcript is noted in the audit log.

> None of this is legal advice: recording laws differ by country and state. Check the laws where you and your callers are.

CHAPTER 38

Recording consent

In the web interface: `/settings`

Some places let a call be recorded when one person on it knows; others need everyone to be told, or to agree. In Settings, **Call recording**, choose how people on a recorded call are told:

- **No notice**.
- **Announce**: as the call is answered, both sides hear the notice (record it in the same place, for example "This call is recorded for training and quality"), and recording starts only after it has played. Until one is recorded, three beeps are played, and Health says so.

- **Beep**: a short tone to both sides every 15 seconds while the call is recorded.

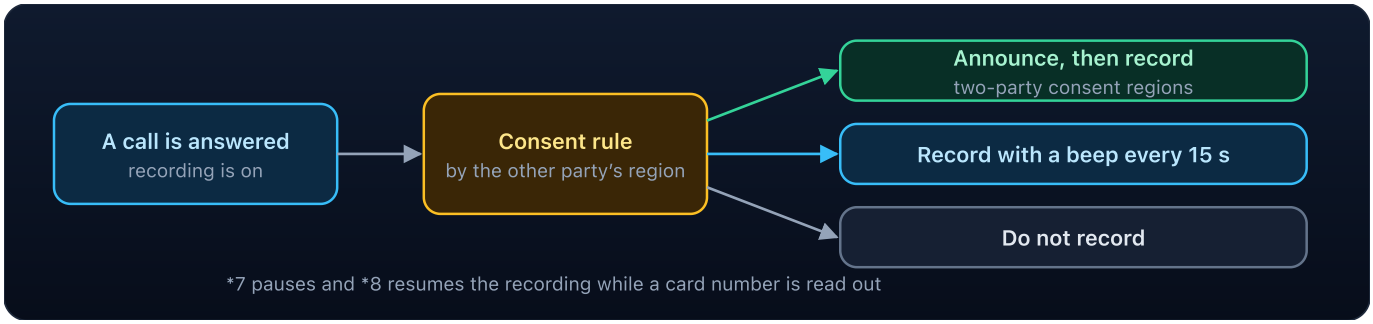


Figure 40. Recording with consent: announce, beep, or not at all

Rules pick by the outside number (the caller on calls in, the number dialled on calls out), longest match first: for example numbers starting 1415 (California) **Announce**, 49 (Germany) **Never record**, everything else as above. A rule's **No notice** records without one; **Never record** leaves those calls out of recording altogether. Calls between extensions follow the main setting.

Recording can also be paused during a call with the pause code (card payments). None of this is legal advice: check the laws where you and your callers are.

CHAPTER 39

Call quality

In the web interface: `/quality`

How calls sounded over the last day, week, month or 90 days, from the sound BitVoice itself received on each side of each call. Phones are grouped by their site (chapter 7) (or "no site"), and calls from outside by the provider's line they came in on.

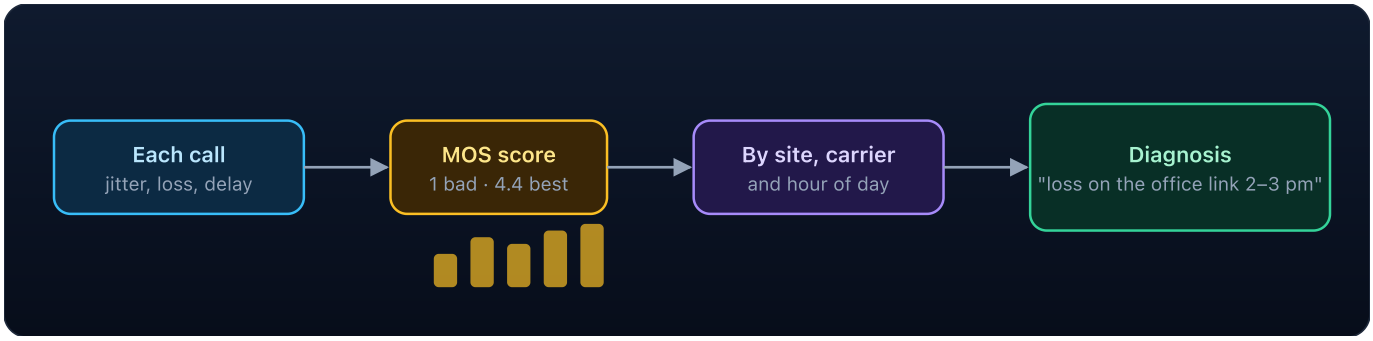


Figure 41. Call quality: from packets to a plain diagnosis

Each place shows its calls, its average score (MOS, out of 5: above 4 is good, under 3.6 sounds poor), the share of sound packets that never arrived, and jitter (uneven delay). Below that, BitVoice says what is wrong in plain words, for example "Packet loss on Head office's network: 3.2% of the sound never

arrived, mostly between 14:00 and 16:00", with what to check. The bar chart shows the sound lost by hour of the day.

If every place loses sound at once, the cause is what they share: the internet link or the provider. Places with fewer than three calls are not judged. Per-call quality is also in Call history (chapter 33) (the Quality column) and, while calls are live, on Live calls (chapter 32).

CHAPTER 40

Costs and budgets

In the web interface: `/costs`

BitVoice prices each call from what your provider charges, and shows the month's spend by department, person and trunk, with the most expensive calls.

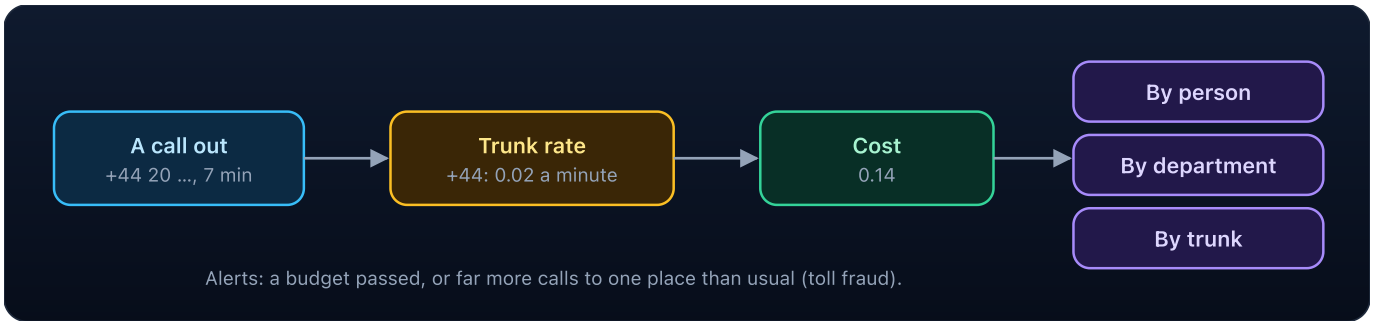


Figure 42. Costs: every call priced, budgets watched

Prices

On Trunks (chapter 4), open a trunk and its **Call prices**. Each price covers numbers that start with some digits, as the number is sent to the provider (digits only, so 44 for the UK, 1900 for US premium rate); the longest match wins, and a price with no digits covers every other number. Give the price per minute, any charge per answered call, the billing step (60 seconds for whole minutes, 6 or 1 for finer) and any minimum. **Calls in, per minute** is for toll-free numbers you pay to receive. **What would it cost** checks a number before anyone dials it. The currency is set on Costs.

A call's cost is fixed when it ends, with the prices of the time; changing a price does not change past calls. Costs says how many answered calls out had no price.

Departments and budgets

Give extensions a **Department** (in the extension). On Costs, add a budget for every call or for one department: an alert appears in BitVoice, and is emailed to the budget's address and the alert address, at

80% and again at 100% of the month.

The alarm

Calls out cost more than this in an hour raises an alarm when outgoing calls pass that amount within any hour. A sudden run of expensive calls is the usual sign of toll fraud (a stolen phone password): check Call history, then disable the extension or the trunk.

CHAPTER 41

Caller ID reputation and branded calls

In the web interface: `/reputation`

Phone companies run spam filters that can label a business number "Spam likely", or block it, and then hardly anyone answers. **Caller ID reputation** watches each number you call out from, from your own calls, and says in plain words how it is doing:

- **Answered**: the share of calls answered, and what it was the period before. A fall by half, or under 20%, is the usual sign of a new label.
- **Hung up at once**: answered calls that ended within 10 seconds.
- **Refused as spam**: calls a carrier's filter refused (SIP 608) or that the person marked unwanted (607). One 608 means the number is labelled.
- Heavy use: more than about 150 calls a day from one number is what robocallers do, and filters notice.

Numbers with fewer than 20 calls are shown as too few to judge.

Your outgoing numbers gives each number a **name** (sent as the caller's name), a **reason** for calling and a **logo** (an https:// image). On a trunk with **Send branded caller ID** on, the logo and reason go out as rich call data (RFC 9796 Call-Info); carriers that support branded calling show them on the called phone. Most phones show the name from the caller-name (CNAM) database instead, which your provider sets: ask them. **Watch** puts a warning on Health when a watched number starts to look labelled.

The page also says what to do when a number is labelled.

CHAPTER 42

Spam protection and STIR/SHAKEN

In the web interface: `/settings`

In the US and Canada, the provider a call starts with signs its caller ID (**STIR/SHAKEN**): **A** when it knows the caller may use the number, **B** when it knows the caller but not the number, **C** when it only passed the call on. Your provider either tells BitVoice its verdict (verstat) or passes the signed Identity header on, which BitVoice then checks itself: the signature, that it is under a minute old, and that it names the numbers of this call. The verdict shows as **Verified**, **Partly verified**, **Unverified** or **Not verified** on Live calls and in Call history.

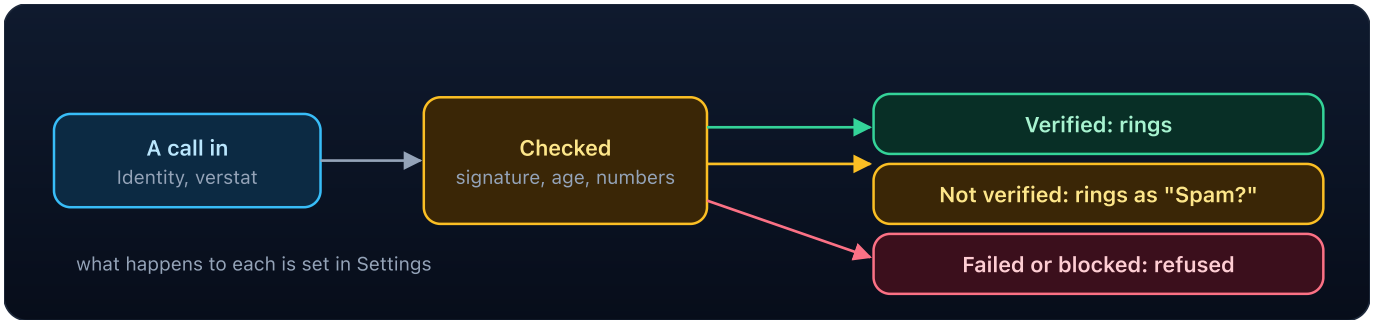


Figure 43. Spam protection: the caller ID is checked before it rings

In Settings, **Spam protection**, choose what happens when a caller ID fails verification (ring, ring labelled, or refuse), when it is not verified at all, and when the number is withheld. A labelled call rings with **Spam?** (or your own label) before the caller's name. **Blocked numbers** are refused; an entry ending in * blocks a range (+1900* is every US premium-rate number). **Always allowed** numbers are never refused or labelled. In Call history, the block button beside a caller adds them to the list; refused calls are listed as **Blocked**.

Within an hour of an emergency call from any extension, no call is refused (blocked ones ring labelled instead), in case it is the emergency service calling back.

To check certificates against the STI authorities, paste their certificates under **STI certificate authorities**; without them, BitVoice checks that the call was signed with the certificate it names but not who issued that certificate.

PART 5

Running BitVoice

Keeping it safe and running: administrators and sign-in, security, changes and updates, several servers, hosted organisations, codecs, Teams, importing, automation and the tools for finding faults.

CHAPTER 43

Administrators

In the web interface: `/users`

Administrators are the people who sign in to this web interface. Staff who only look after their own calls use the staff portal instead and need no account here (The staff portal (chapter 21)).

Roles

Role	Can
Administrator	change anything, play recordings, see every secret field's existence (never its value)
Viewer	see everything, change nothing; secrets are hidden

A change of role, or of password, signs the person out everywhere at once.

Two-step sign-in

Each person turns on two-step sign-in for themselves: **Set up** shows a key for an authenticator app; entering the code it shows turns it on. From then on, sign-in asks for a code as well as the password. Turning it off needs the password.

> Turn on two-step sign-in for every administrator of a system reachable from the internet.

Single sign-on

With single sign-on (Settings), people sign in with the organisation's account (Microsoft Entra, Google, Okta, any OpenID Connect provider), and an account is made for them the first time. See Single sign-on (chapter 44).

API tokens

Programs (Terraform, scripts) use API tokens instead of a password: made on this page, shown once, and revoked at any time. Each token is an administrator or a viewer.

CHAPTER 44

Single sign-on

Single sign-on lets people sign in to BitVoice with their work account, through any OpenID Connect provider: Microsoft Entra ID, Google, Okta, Keycloak and others. Leaving the organisation then removes their access here too.

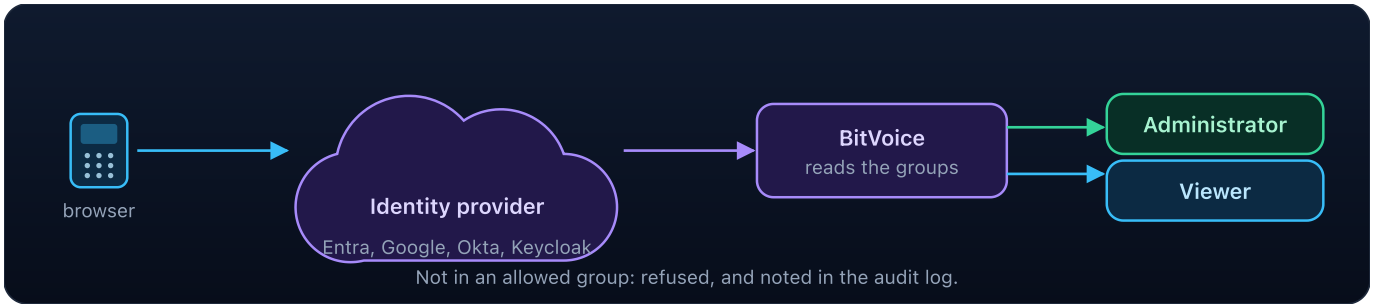


Figure 44. Single sign-on: the work account decides who gets in

Setting it up

In Settings (chapter 9), **Single sign-on**:

1. Register an application with the provider, with the **Redirect address** shown here.
2. **Fill in for a provider** sets the **Issuer** for the common ones; copy the **Client ID** and **Client secret** from the provider.
3. Name the provider's **Administrator groups** and **Viewer groups**. **Groups claim** is where the provider puts them (`groups` unless set).
4. Optionally change the **Button text** on the sign-in page.
5. Switch on **Use single sign-on**.

How it works

- The account here is made at the first sign-in; the groups set the role at every sign-in, so moving someone between groups at the provider changes their role here.
- Someone in neither kind of group is refused, and the refusal is in the audit log.
- With **Also allow passwords** off, passwords work only on this computer, so a provider outage never locks you out.

> Two-step sign-in is then the provider's job: switch it on there.

CHAPTER 45

Security

In the web interface: `/security`

A phone system on the internet is attacked within hours: scanners try passwords on every SIP port, and a broken-into system runs up calls to premium numbers. BitVoice is built to stand up to that by default, in layers.

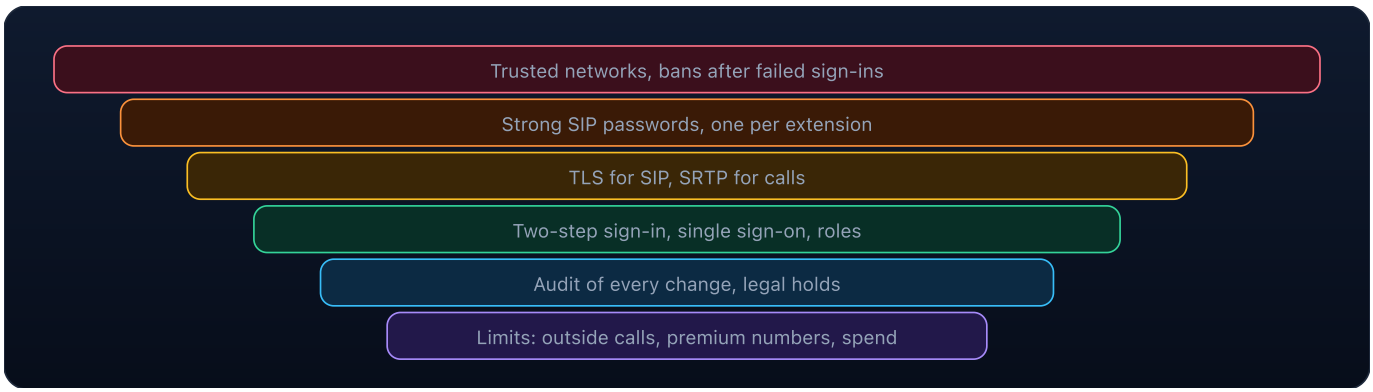


Figure 45. Layers of protection

Against password guessing

- Addresses that keep failing to sign in are banned for a while (Settings, Protection): by default ten failures in a minute ban the address for an hour. Banned addresses are listed on Security (chapter 45) and can be unbanned.
- **Trusted networks** are never banned: your office network.
- SIP passwords are generated, at least 12 characters, one per extension. A phone signing in as one extension can never register another.
- A scanner that is banned gets no answer at all, so it learns nothing.

Against fraud

- Extensions can call outside numbers only when allowed.
- Premium-rate numbers are refused unless a route names them.
- Costs (chapter 40) alerts on budgets and on runaway spend: many calls to one country in an hour.
- Each extension may have at most four calls at once (Settings, Protection).

Encryption

- **TLS** encrypts signalling (port 5061), so passwords and numbers are not readable on the network.
- **SRTP** encrypts the sound of calls; the browser phone and apps always use it.
- The web interface has HTTPS on port 8443 with its own certificate, or one you install.

Who can do what

- Administrators change everything; viewers see everything but change nothing; staff use the portal for their own calls only.
- Turn on **two-step sign-in** for every administrator (Administrators (chapter 43)): a code from an authenticator app.
- Single sign-on (chapter 44) signs people in through your identity provider.
- The **audit log** records every administrative action and who did it, and cannot be edited.

CHAPTER 46

Changes, backups and import

In the web interface: `/changes`

Every saved change is a revision: numbered, with who made it, when, and a one-line summary. Nothing in the configuration is ever lost.



Figure 46. Every change is a revision you can compare and undo

Comparing and restoring

- **What changed** shows the difference between a revision and the one before, field by field.
- **Restore** shows what restoring would change and asks first. Restoring is itself a new revision, so it can be undone too.

Export and import

- **Export** downloads the whole configuration as one file (secrets left out unless you ask for them).
- **Import** previews a file's changes before applying them.
- Import (chapter 53) also reads FreePBX and Asterisk configurations, 3CX user lists, and the CSV exports of Cisco Unified CM, Avaya and Mitel.
- Archive (chapter 56) copies backups, recordings and voicemail to Amazon S3 or Azure.

Changes made at the same time

Two administrators editing the same thing at once cannot overwrite each other: the second save is refused with a message to reload, and nothing is lost.

CHAPTER 47

Updates

In the web interface: `/updates`

Updating BitVoice is one click, and safe: it checks the new version, installs it beside the old one, and goes back on its own if the new one does not stay up.

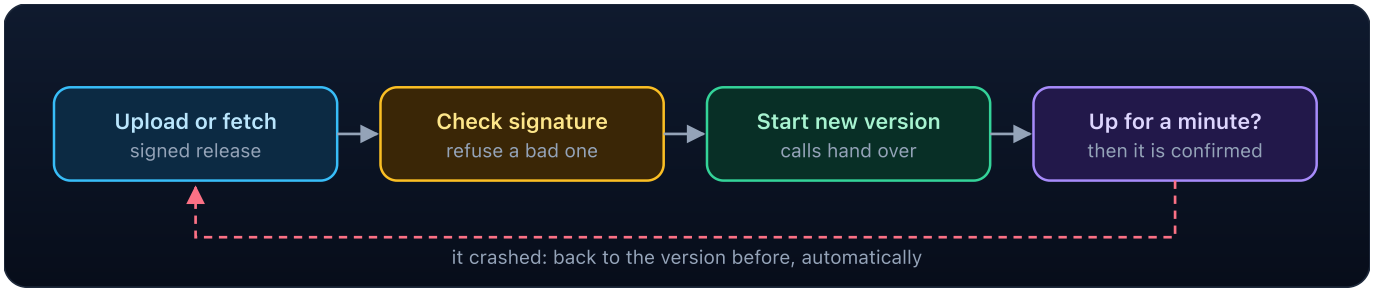


Figure 47. Updates: install, check, and fall back on their own

Installing an update

1. On Updates (chapter 47), upload the new BitVoice program and its `.sig` file.
2. Press **Install**. By default BitVoice waits until no calls are in progress; you can also install at once.

3. BitVoice restarts on the new version. With several servers, calls in progress are handed to another server and carry on.

Going back

The version you left is kept: **Go back** returns to it in one click. A new version that fails to start twice, or does not stay up for a minute, is replaced by the previous one by itself.

> Release builds install only versions signed by BioDream, so a forged program cannot be installed through this page.

CHAPTER 48

Several servers as one phone system

In the web interface: `/servers`

Two or more BitVoice servers that share one PostgreSQL database work as one phone system:

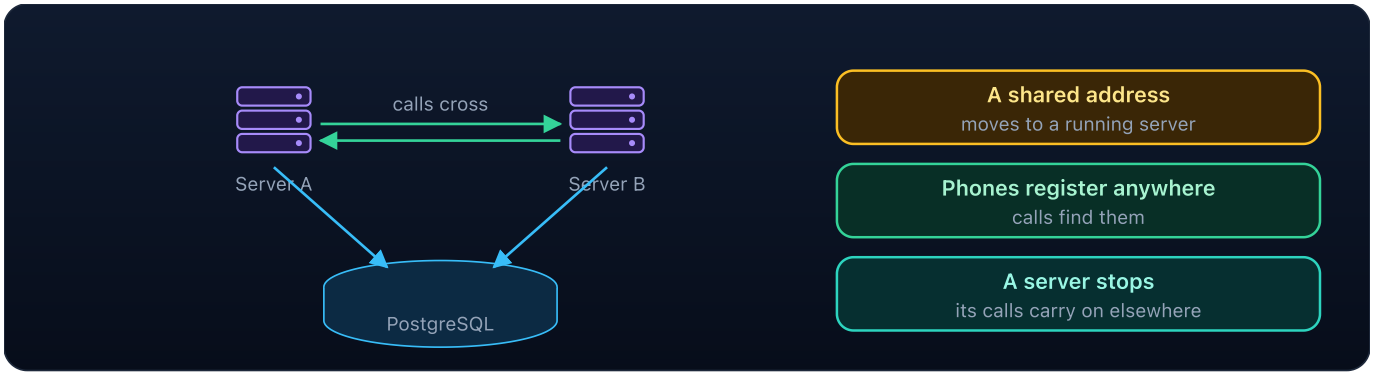


Figure 48. Several servers as one phone system

- A call reaches a phone **whichever server it is registered with**: ring groups, queues, transfers to an extension and calls from provider lines all ring phones on every server.
- **Settings** changed on one server apply on all within five seconds; users, contacts, call history, recordings' transcripts and texts are shared.
- **Busy lamps** (BLF) show phones on every server ringing and busy; **parking** slots are shared, so a call parked on one server is picked up from a phone on another; **paging** reaches phones on every server; **texts** reach desk phones wherever they are registered.
- A queue agent on a call on one server counts as busy on the others, and a provider line's channel limit counts calls on every server.
- **Live calls** on each server lists the calls on the others too (read-only: hang up or listen in on the server that has the call).

- The call record is written once, by the server that has the caller, and a call is recorded there too.

Setting it up

1. Move the first server's data into PostgreSQL: Settings, **Database**.
2. Start each other server with the same database: `bitvoice -database postgres://…` (or the BITVOICE_DATABASE environment variable). It joins at once; **Servers** lists it.
3. On **Servers**, check where each server is reached (its address and SIP port on the network the servers share), and give each a name.
4. Tell phones how to find the servers (below).
5. Provider lines (below).

How phones find the servers. Phones BitVoice sets up (Yealink, Polycom, Grandstream) are given another running server as their backup by themselves. For the others, Snom and phones set up by hand, use one of these on **Servers**:

- **A DNS name for every server**: a name such as pbx.example.com with SRV records (_sip._udp, _sip._tcp, _sips._tcp) listing each server. Phones BitVoice sets up are given the name, and Servers checks the records against the running servers, lists what is missing and gives the records to create, ready to copy.
- **A shared address**: an unused address on the servers' network, such as 192.168.1.50/24. One running server holds it at a time and tells the network at once (gratuitous ARP); if it stops, the next server takes the address within seconds, carries on its calls and answers its phones at the same address, without them registering again. It needs Linux and the NET_ADMIN and NET_RAW permissions (in a container: `--cap-add NET_ADMIN --cap-add NET_RAW`), and the servers on one network. Restart each server after setting it.

Provider lines. In each trunk, **Shared between servers** says how:

- **Each server uses it**: every server registers and places its own calls. For providers that allow several registrations on one account, or none.
- **One server at a time**: one server registers and carries the line; the others show it as **Standby, held by** that server and send their calls out through it. If it stops, the next server registers within seconds. For providers that allow one registration per account.

When a server stops

- **Calls carry on.** Each server keeps a note of its calls in the database; when one stops, a running server picks up each call within seconds, moves the sound to itself and tells both ends (a re-INVITE). There is a gap of a few seconds in the sound. A planned stop (an upgrade, a restart) hands calls over at once, with no gap.
- Its phones carry on through their backup server, the DNS name or the shared address.

- Health and **Servers** show it as stopped; when it comes back it takes nothing back from the server that took over.

Between servers. The servers reach each other at their SIP port over TCP, or TLS when they have a certificate (each accepts only the certificate the other published, and the sound between them is then encrypted with SRTP), and send sound to each other's media ports (UDP, the range in Settings). Calls between servers are signed with a secret they share in the database, and only accepted from a running member at its own address. A firewall between them must let those ports through.

Try it first: the lab. BitVoice comes with a lab that runs two servers, their database and test phones on one computer with Docker, each on an address of its own: `deploy/lab/run.sh up` builds and starts it and sets the shared address; `run.sh call` calls between phones on different servers; `run.sh failover` kills the server carrying a call through the shared address and checks the call carries on (the sound comes back in about five seconds); `run.sh impaired` calls from a phone on a bad link (delay, jitter and 5% loss) and checks the call quality BitVoice reports; `run.sh down` removes it. The servers' pages are at http://127.0.0.1:18501 and :18502 (user admin).

CHAPTER 49

Where the data is kept (SQLite or PostgreSQL)

In the web interface: `/settings`

BitVoice keeps its settings and their saved versions, users, contacts, call records, the audit log, texts, notes and transcripts in a database. Recordings and voicemail stay as sound files in the data folder.

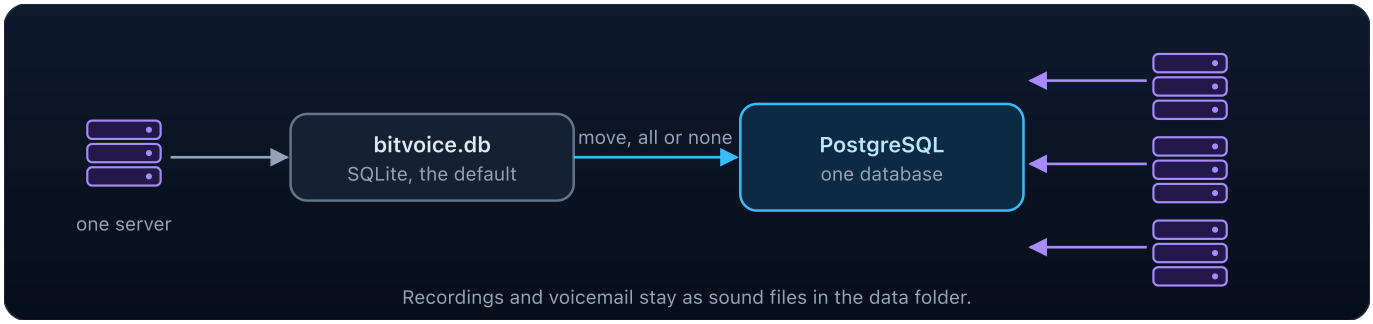


Figure 49. Where the data is: one file, or PostgreSQL

This server's own file (SQLite, the default): a single file, `bitvoice.db` in the data folder, with nothing to install. Back up the data folder and you have everything.

PostgreSQL: for several BitVoice servers sharing one set of data, or for reports in tools such as Power BI, Grafana or Metabase. In Settings, **Database**, **Move the data into PostgreSQL**: make an empty database and a user who owns it, enter its address (postgres://user:password@host:5432/database), press **Test**, then **Copy the data and restart on PostgreSQL**. Everything is copied in one go (all of it or none of it)

and BitVoice restarts on it; the old file stays where it was. Moving back into the server's own file works the same way. Moves are refused while calls are in progress.

Servers sharing a PostgreSQL database see each other's settings changes within five seconds. Each still answers its own phones and calls; for a second server to take over a first one's phones, the phones need both addresses (or one address that moves).

The database can also be named when BitVoice starts, with `-database postgres://…` or the BITVOICE_DATABASE environment variable; it is then changed there, not in Settings.

Older installations: on the first start of this version, BitVoice copies its old JSON files into the database and moves them into `pre-database` in the data folder.

CHAPTER 50

Hosting other organisations

In the web interface: `/tenants`

One BitVoice can host several organisations. Each one is a phone system of its own: its own extensions, numbers, trunks, routes, administrators, voicemail, recordings and call history. No organisation can see another's. They share only this server's SIP port, IAX2 port and media ports.

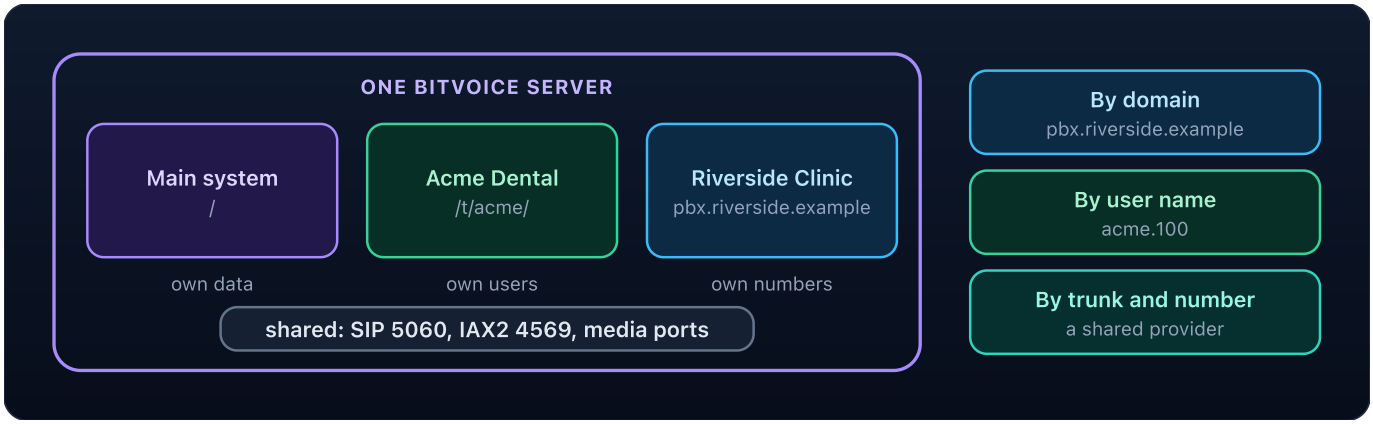


Figure 50. Hosted organisations: one server, separate phone systems

When to use it

- A provider, or an IT company, running phone systems for many small customers on one server.
- A group of offices or clinics that want separate systems, administered separately, on shared hardware.
- Several answering points sharing one system, each keeping its own call-takers and records.

The main system is an organisation too: the one this server was set up as. Its administrators manage the others.

Adding an organisation

1. Open Organisations (chapter 50) and press **Add organisation**.
2. Give its **name** and a **short name** (lowercase letters, digits and dashes). The short name is part of its web address, `/t/acme/`, and of its phones' sign-ins, `acme.100`. It cannot be changed later.
3. Optionally give its **own domain**, such as `pbx.acme.example`, pointed at this server. Phones register to that domain, and its administrators open it in a browser.
4. Set its **first administrator**: a username and a password of at least 12 characters, and its country.
5. Optionally set **limits**.

The organisation starts at once with an empty phone system, ready for its administrator to set up as described in Getting started (chapter 1).

How calls find the right organisation

A request	goes to
from a phone registering to `pbx.acme.example`	the organisation with that domain
from a phone signing in as `acme.100`	Acme, extension 100
from a provider, for a number	the organisation whose trunk it comes from and whose incoming route has that number
anything else	the main system

So two organisations can both have extension 100, and several can share one provider account with different numbers.

Where its data is

On SQLite, each organisation has its own folder under `tenants/` in the data folder, with its own database file. On PostgreSQL, each has its own schema in the same database (`tenant_acme`). Backups, archives and exports are per organisation.

Limits

Limit	What happens at it
Extensions	adding one more is refused with a message
Calls at once	new calls are refused (503) until one ends
Storage	new recordings and voicemail are not kept
AI requests a day	transcripts, summaries and the AI receptionist stop until midnight UTC

0 is no limit. Emergency calls are never refused: an organisation at its call limit, or suspended, can still call 911 and be called back.

Opening an organisation

Open signs you in to its web interface as the provider, without its own account; it shows in its audit log as the provider. Its administrators sign in at its own address with their own accounts. Each organisation has its own sign-in cookie, so being signed in to one never signs anyone in to another.

Organisations never see this server's own screens: updates, the database, the other servers, the certificate, the SIP trace (which shows every organisation's messages), the call generator or this list.

Suspending and removing

Suspended stops sign-ins and every call except emergency calls; nothing is deleted. **Remove** stops the organisation and moves its folder aside (`tenants/.removed-<name>-<time>`), so a mistake can be undone by hand.

CHAPTER 51

Codecs and sound

A codec is how sound is turned into data on a call. Phones, browsers and carriers each support some; BitVoice converts between any two of its codecs when two parties differ, so everyone can talk to everyone. When both use the same codec the sound passes through untouched.

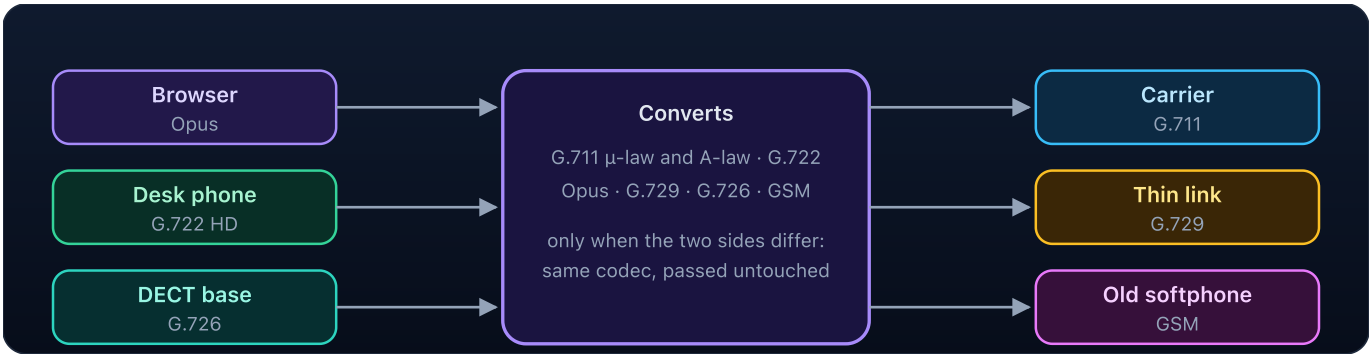


Figure 51. Codecs: each side speaks its own, BitVoice converts in between

The codecs

Codec	Sound	Bandwidth	Used by
G.722	HD (wideband)	64 kbit/s	most desk phones: the best choice between them
G.711 µ-law	telephone	64 kbit/s	North America and Japan; every carrier
G.711 A-law	telephone	64 kbit/s	the rest of the world
Opus	HD, copes with loss	24 kbit/s	browsers and the mobile apps
G.729	telephone, a little rougher	8 kbit/s	thin links, some carriers
G.726 (32 kbit/s)	near telephone	32 kbit/s	DECT bases, older gateways
G.726 (16 kbit/s)	rough	16 kbit/s	very thin links
GSM	rougher	13 kbit/s	older softphones and gateways

Settings lists the codecs phones may use, best first; each trunk may have its own list. Analog adapter ports always use G.711, which faxes, modems and alarm panels need.

> G.726 comes in two packings: RFC 3551's (G726-32) and the older AAL2 order. Most equipment uses the first; choose **G.726 (AAL2 order)** only for equipment that sounds garbled with it.

What converting costs

Converting takes processor time, only on calls where the two sides differ. Per call, about 0.3 ms of one core every 20 ms for Opus and 0.12 ms for G.729 (some 60 and 170 converted calls per core); G.726 and GSM cost less than G.729; G.711 and G.722 almost nothing. G.729 is sent without Annex B (no silence frames).

Sound clean-up

Sound clean-up (Settings) turns down steady background noise and evens out levels, for voicemail and, if you like, during calls.

CHAPTER 52

Microsoft Teams (Direct Routing)

In the web interface: `/trunks`

BitVoice can be the session border controller (SBC) for a Microsoft Teams tenant, so Teams users call BitVoice extensions and your phone lines, and BitVoice rings Teams users.

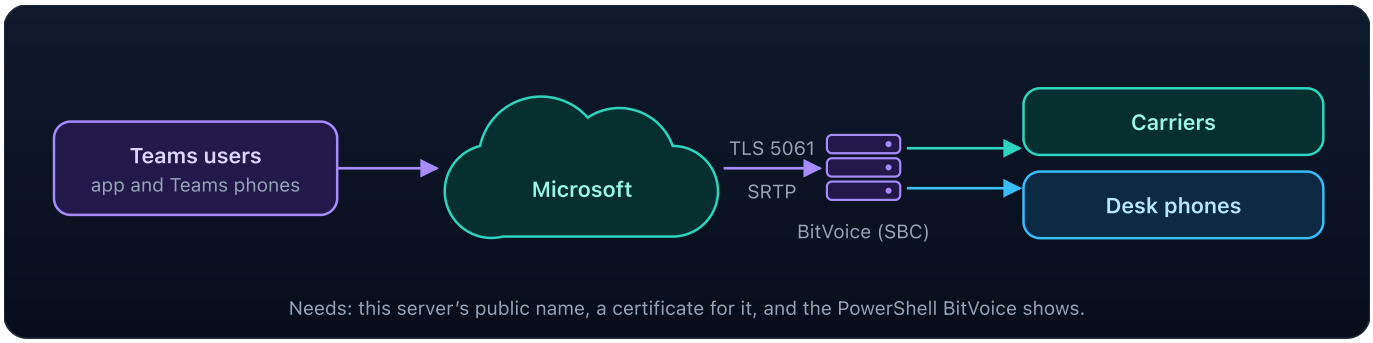


Figure 52. Microsoft Teams Direct Routing: BitVoice is the SBC

What you need: a public name for this server in a domain your Microsoft 365 tenant has verified (sbc.example.com), port 5061 open both ways to Microsoft's networks 52.112.0.0/14 and 52.120.0.0/14, and a certificate for that name from a public authority. Teams users need a Phone System licence.

Steps

1. Settings, **Certificate**: install the certificate and its private key (PEM). It is used at once, for this interface and for SIP over TLS, and BitVoice presents it to Microsoft.
2. Trunks, **Add**, start from **Microsoft Teams**: enter the server's public name. The drawer says whether the certificate covers it.

3. Copy the **PowerShell** the drawer shows and run it as a Teams administrator: it adds the SBC (New-CsOnlinePSTNGateway), a voice route and policy, and a dial plan rule so short extension numbers reach BitVoice as dialled; then give each Teams user the policy and their number.
4. To ring Teams users from BitVoice, add an outbound route for their numbers through the Teams trunk.

BitVoice sends Microsoft OPTIONS every minute with the SBC name in Contact, as Microsoft requires, and tries sip, sip2 and sip3.pstnhub.microsoft.com in turn. Media is always encrypted (SRTP). Health warns when the certificate does not cover the name, is self-signed or ends soon, and when Microsoft stops answering.

This follows Microsoft's published SBC requirements and is tested against a simulated Microsoft proxy; BitVoice is not on Microsoft's list of certified SBCs.

CHAPTER 53

Import

In the web interface: `/import`

Brings extensions, ring groups, queues and trunks over from FreePBX, Asterisk or 3CX; and people, hunt and ring groups, SIP trunks, incoming numbers and dial patterns from Cisco, Avaya and Mitel. Upload the exported files, several at once (a .zip or .tar of them works too); BitVoice shows exactly what would change before anything is applied, and the import becomes a revision you can undo on Changes (chapter 46).

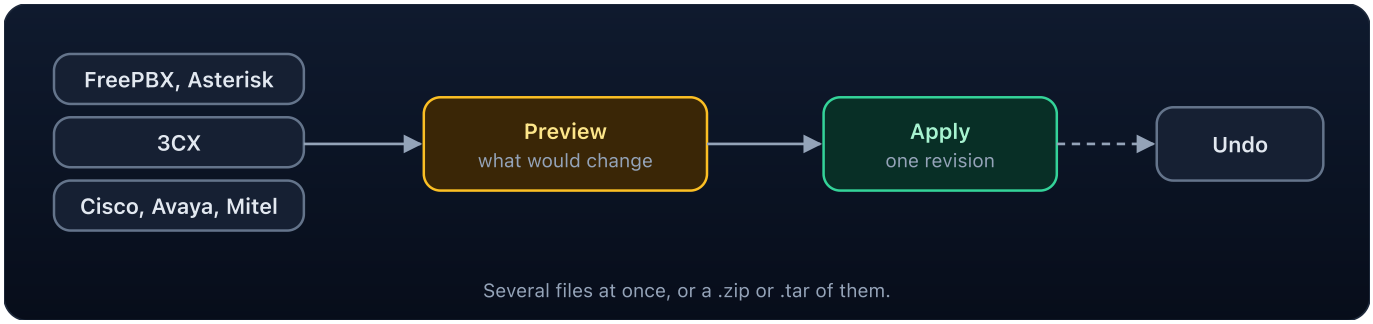


Figure 53. Import: another system’s export, previewed before anything changes

Where each export is:

- **FreePBX:** Admin → Bulk Handler → Export → Extensions (CSV).
- **Asterisk:** pjsip.conf or sip.conf.
- **3CX:** Users → Export (CSV).
- **Cisco Unified CM:** Bulk Administration → Phones → Export Phones, all details; or Users → Export Users. The phones' first line becomes the extension, with the line's alerting name or label as its

name.

- **Avaya IP Office:** Manager → File → Import/Export → Export, Users.
- **Avaya Communication Manager:** `list station` saved as CSV; virtual stations and VDNs are left out.
- **Mitel MiVoice Business:** System Data Management → User and Device Configuration → Export.

Groups and routes. Upload them with the people file, so each group's members are found by number or by name:

- **Cisco Unified CM:** Bulk Administration → Import/Export → Export, with Line Group, Hunt List, Hunt Pilot, SIP Trunk and Route Pattern ticked; upload the .tar as it comes. Each hunt pilot becomes a ring group with the members of its hunt list's line groups, Top Down and Circular as one after another, Broadcast as everyone at once, the RNA reversion timeout as the ring time. Route patterns become outbound routes: `9.[2-9]XXXXXX` with PreDot is pattern `9[2-9]XXXXXX` dropping the 9; `!` becomes a final dot. Patterns using `@` (the national plan) are listed for writing by hand.
- **Avaya IP Office:** export Hunt Group as well as Users (both are CSV with no header row). Group mode is everyone at once; Hunt, Rotary and Longest Waiting ring one after another.
- **Avaya Communication Manager:** `list hunt-group` saved as CSV, with the members in a Members column or a separate members file.
- **Mitel MiVoice Business:** Ring Groups and Ring Group Assignment (Cascade is one after another, Ring All everyone at once), and Hunt Groups.

Trunks come in switched off and without passwords, as exports do not carry them. Routes go out on the imported trunk of the same name; a route whose trunk is not here is left out and said. Rotating and longest-idle ringing is closer to a queue than a ring group: the preview says so, so you can make it one.

These exports hold no SIP passwords, so each extension gets a new one; voicemail PINs, names and email addresses come over when the export has them. Cisco, Avaya H.323/DCP and Mitel phones run their maker's own software and cannot be set up by BitVoice: the import says how many were listed. Cisco phones can be re-flashed with Cisco's multiplatform (SIP) firmware; Avaya J100 and Mitel 6800/6900 phones can register to BitVoice by hand in SIP mode.

CHAPTER 54

Configuration as code

Everything the screens do goes through BitVoice's REST API, so anything can be scripted, kept in Git, or managed with Terraform. The API is described for tools at `/api/openapi.json` (/api/openapi.json).

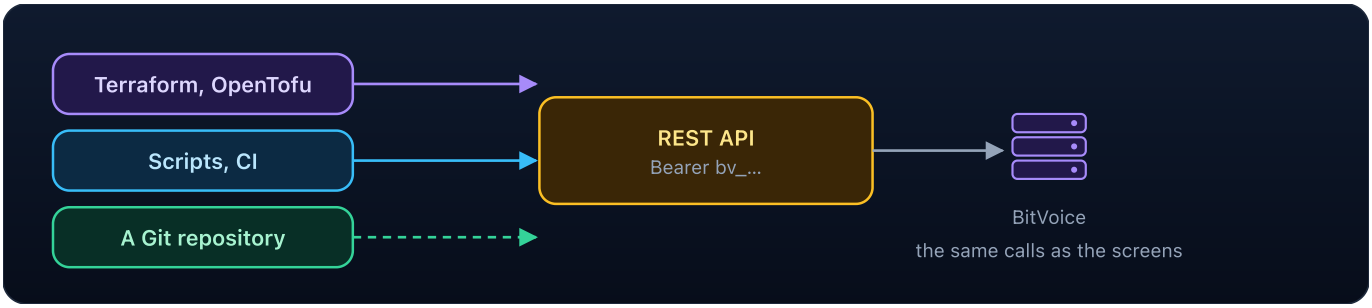


Figure 54. Configuration as code: the API, Terraform and Git

API tokens

Made on Administrators (chapter 43): one per program, read only or able to change everything. Send it as `Authorization: Bearer bv_...`. It is shown once; revoke it when no longer needed. A token's changes show in Changes and the audit log under its name.

```
curl -H "Authorization: Bearer bv_..." https://pbx.example.com/api/extensions
```

Terraform or OpenTofu

The repository's `deploy/terraform` shows extensions, ring groups and routes managed with the generic REST provider: plan shows what would change, apply makes the change.

Configuration from Git

In Settings (chapter 9), **Configuration from Git**, BitVoice reads the configuration from a file in a repository every few minutes and either shows what differs or makes the changes. Start the file with **Export for Git**, which leaves out passwords; BitVoice keeps its own, so secrets never go into the repository.

> With BitVoice making the changes, a change made in the screens is undone at the next read if the file says otherwise: choose one place to make changes.

CHAPTER 55

Asterisk compatibility (AMI and ARI)

Tools written for Asterisk can control BitVoice: CRMs, wallboards, click-to-dial buttons, call recorders' controllers. Switch it on in Settings (chapter 9), **Asterisk compatibility**. Tools see calls as channels named like `PJSIP/100-...`, as on Asterisk.



Figure 55. Asterisk compatibility: tools written for Asterisk drive BitVoice

AMI

The manager interface on TCP port 5038 (**AMI listens on**: this computer only by default). Supported:

Kind	Names
Actions	Login (plain or MD5 challenge), Ping, Events, CoreShowChannels, Status, PJSIPShowEndpoints, ExtensionState, QueueStatus, Originate, Redirect, Hangup, Logoff
Events	Newchannel, Newstate, Hangup

ARI

At `/ari/`, with basic authentication or `?api_key=user:secret`:

Request	What it does
`GET /ari/channels`	the calls in progress
`POST /ari/channels?endpoint=PJSIP/100&extension=5551234`	ring 100, then call the number
`DELETE /ari/channels/{id}`	hang up
`POST /ari/channels/{id}/redirect?endpoint=PJSIP/102`	send the call to 102
`/ari/endpoints`, `/ari/bridges`, `/ari/asterisk/info`	phones, bridges, version
`/ari/events`	a WebSocket of events

Stasis media applications are not offered: use Scripts (chapter 15) for call logic.

Logins

Each login has a **Username** and **Secret**. Logins are **read only** unless marked **Can change calls**. Keep AMI on this computer unless the tool runs elsewhere: AMI is plain, unencrypted TCP.

CHAPTER 56

Archive

In the web interface: `/archive`

The archive keeps copies of recordings, voicemail and the configuration somewhere else: an Amazon S3 bucket (or any S3-compatible store) or an Azure Blob Storage container. If this server's disk is lost, the archive still has them.

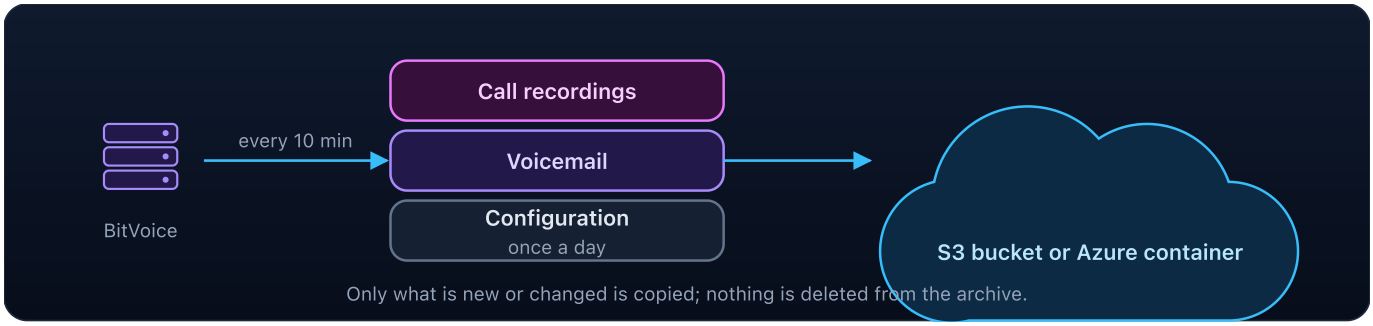


Figure 56. The archive: copies kept somewhere else

Setting it up

1. On Archive (chapter 56), choose **Where to archive**: Amazon S3 or Azure.
2. For S3: the **Bucket**, its **Region**, an optional **Folder in the bucket** and **Endpoint** (for S3-compatible stores), and an **Access key ID** and **Secret access key**. **Copy the policy** gives the only permission BitVoice needs, to attach to the user or role.
3. For Azure: a **Container SAS URL** with write permission, and an optional **Folder in the container**.
4. Choose what to copy besides the configuration: **Call recordings**, **Voicemail**, or both.

What happens

Every ten minutes BitVoice copies new recordings and messages that are not in the archive yet, and once a day a dated copy of the configuration (`config/2026-10-13.json`). The page shows the last run, the last success and any error.

> Keys can be references to a key vault instead of the keys themselves (Settings, Key vaults), so they are never stored in BitVoice.

Deleting something here (by retention or by hand) does not delete the archive's copy: set the bucket's or container's own lifecycle rules for how long the archive keeps things.

CHAPTER 57

Legal holds

In the web interface: `/holds`

A legal hold keeps call records, recordings and voicemail that a matter may need, and stops anyone deleting them, until it is released. Use it when a lawyer asks for records to be preserved.

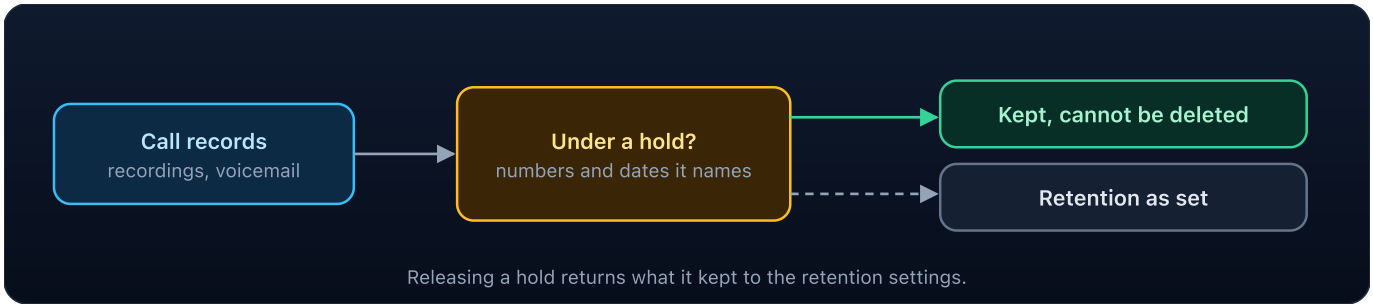


Figure 57. A legal hold: kept, whatever the retention says

Adding a hold

On Legal holds (chapter 57), press **Add**:

Field	What it is
Name	what the matter is called
Reason	why, for the record
Numbers	the extensions and outside numbers it covers (`101, +15551234567`), or **Everything**
From, To	the dates it covers; empty for no limit

While a hold is on

- Retention leaves what it covers alone, whatever its age.
- Nobody can delete a held call, recording or voicemail, administrators included.
- Held calls show a lock in Call history (chapter 33).

Releasing

Release makes what the hold kept subject to the retention settings again: things older than the retention are deleted at the next clean-up. Released holds stay listed, with when and by whom, and adding and releasing holds is in the audit log.

CHAPTER 58

Metrics: Prometheus, OpenTelemetry and Grafana

In the web interface: `/settings`

Prometheus scrapes `/metrics` on the web address. Make an API token under Administrators (a viewer token is enough) and give it to Prometheus as its bearer token:

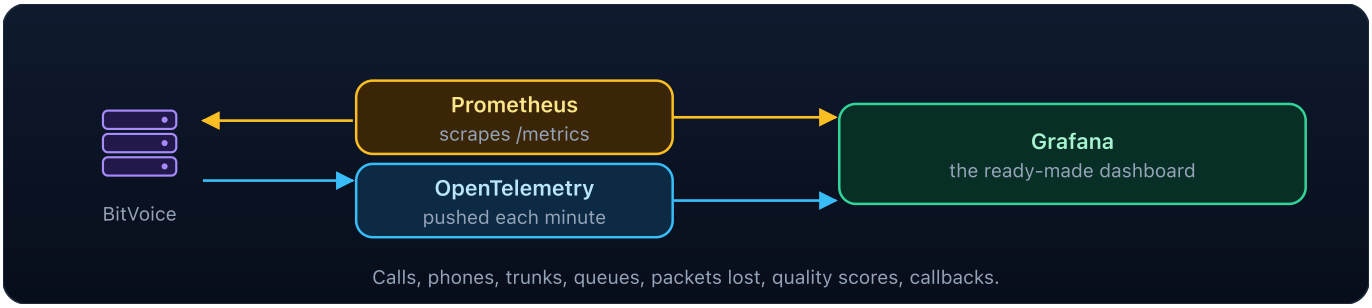


Figure 58. Metrics: Prometheus pulls, OpenTelemetry is pushed

```
scrape_configs:
  - job_name: bitvoice
    scheme: https
    authorization: { credentials: bv_... }
    static_configs: [{ targets: ['pbx.example.com'] }]
```

It reads calls in progress (by state), calls ended (by direction and outcome), talk time, phones registered, each trunk's state and round trip, callers waiting and agents in each queue, sound packets received and lost, a histogram of call quality scores (MOS), and callbacks booked. Counters start again from zero when BitVoice restarts, as Prometheus expects.

OpenTelemetry: in Settings, Metrics, enter a collector's OTLP/HTTP address (and a header such as Authorization, with its value) to have the same measurements pushed every minute (or as often as you set, from 15 seconds). **Push now** checks the saved settings.

Grafana dashboard downloads a ready-made dashboard: import it in Grafana and pick your Prometheus data source.

CHAPTER 59

Call generator

In the web interface: `/loadtest`

The call generator places real calls with real audio into BitVoice from this computer: to see how many calls a server carries and how well, or to check after a change that calls still go where they should.

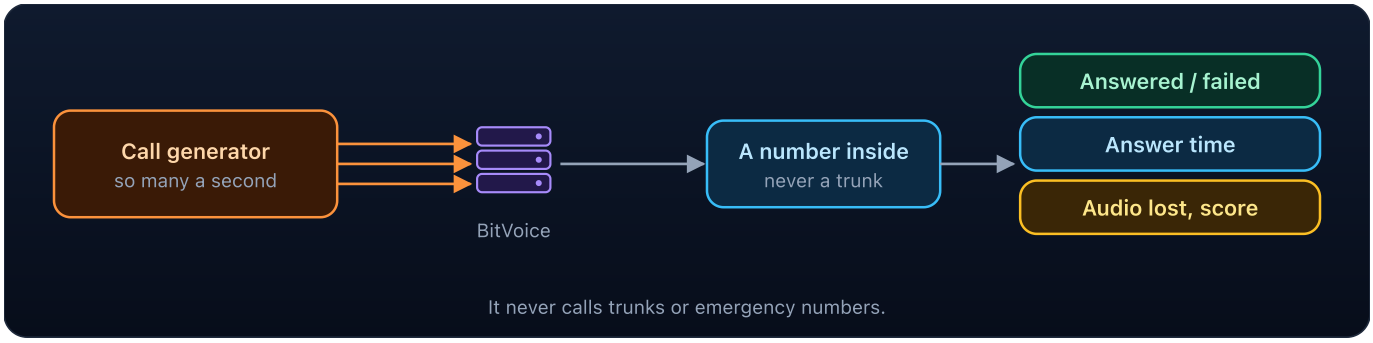


Figure 59. The call generator: real calls, real audio, measured

Running a test

Setting	What it is
Number to call	a number inside BitVoice: an extension, a group, a queue, a menu
Calls come from	the caller number the calls show
Answer as	an extension the generator answers for itself; or nobody, when the number answers by itself (a menu, the echo test)
Calls	how many in all
At once	the most in progress together
New per second	how fast they start
Each lasts (s)	how long each talks

Reading the results

Result	What it says
Answered, Failed	and **Why calls failed**, by reason
Answer time, median and 95%	how long callers waited
Audio lost	the share of sound packets that never arrived
Audio score	an estimate of the sound (MOS): 4 and over is good, under 3.6 is poor
Calls in progress over time	a chart of the load

> It only calls numbers inside BitVoice, never trunks or emergency numbers, so a test can never cost money or reach 911. It is for this server's administrators only.

CHAPTER 60

SIP trace

In the web interface: `/trace`

The SIP trace shows the messages BitVoice sends and receives, live: the first place to look when a phone will not register or a call fails, and what a provider's support desk will ask for.

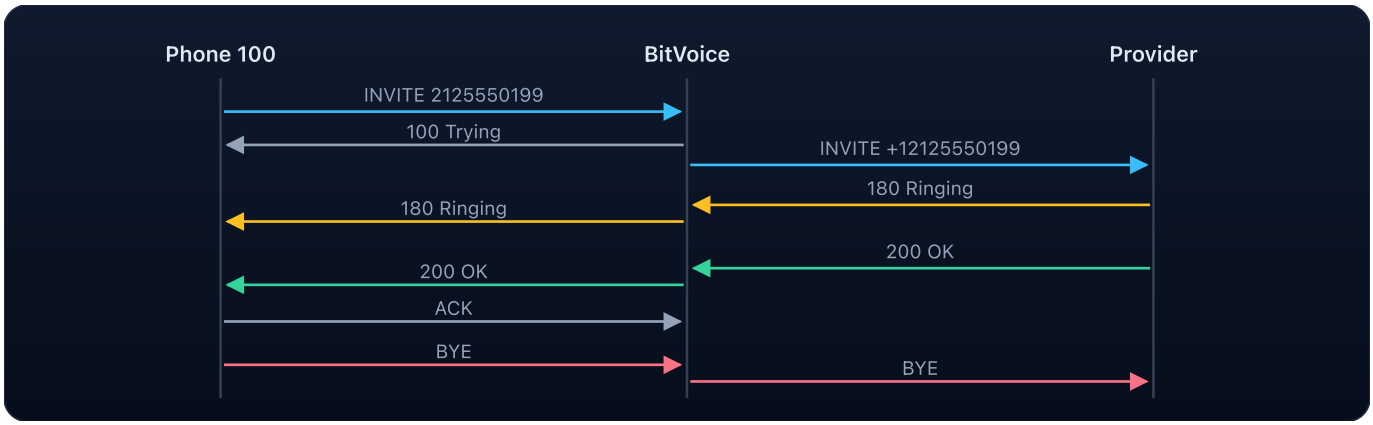


Figure 60. A SIP trace: the messages of one call, in order

Finding a message

BitVoice keeps the last 2,000 messages in memory (nothing is written to disk, and they are gone after a restart). Type part of a number, an address or a Call-ID in the filter. Press **Refresh** for the newest.

Choose a message to see it in full and, beside it, the **call flow**: every message of the same call in order, between the phone, BitVoice and the provider, as in the figure.

Reading a trace

You see	It usually means
REGISTER answered 401, then nothing	the phone does not have the password, or has the wrong one
REGISTER answered 403	the address is banned, or the phone signs in as another extension
INVITE answered 404	no route matches the number dialled (Call routing (chapter 5))
INVITE answered 403 by the provider	the provider refuses the caller ID or the number
486	the other side is busy
488	the two sides have no codec in common (Codecs (chapter 51))
200 OK, then BYE after about 30 seconds	the ACK never arrived: a router in between rewrites or drops messages (switch off its "SIP ALG")
A call connects but nobody hears anything	the sound is blocked: check the public address in Settings, and the media ports in the firewall

> A trace shows phone numbers and sign-in names. Only people signed in to this server see it; hosted organisations never do, since it would show every organisation's messages.

PART 6

Public safety: the lab track

The 911 answering-point profile, the radio dispatch console and NG911 Core Services. They are a lab and design-partner track: built and tested against recorded captures, sample data and stand-ins, but not certified for live 911 or radio dispatch and not yet tried with a live ESInet, P25 system or CAD. They are off the menu until switched on in Settings. Emergency calls from your own phones (Getting started) are not part of this track: they are part of every phone system.

CHAPTER 61

911 calls (the 911 profile)

In the web interface: `/psap`

> **Lab and design-partner track.** Not certified for live 911 or radio dispatch, and not yet tried with a live ESInet, P25 system or CAD. Use it in a lab or in shadow mode beside your current equipment. Switch on **Public-safety profiles (lab)** in Settings (chapter 9) to show it in the menu.

BitVoice answers 911 calls as a PSAP's call-handling system: calls from the emergency services network (the ESInet, NENA i3) arrive with the caller's location and the network's data, the call-taker sees them at once, and CAD is told when the call is answered.

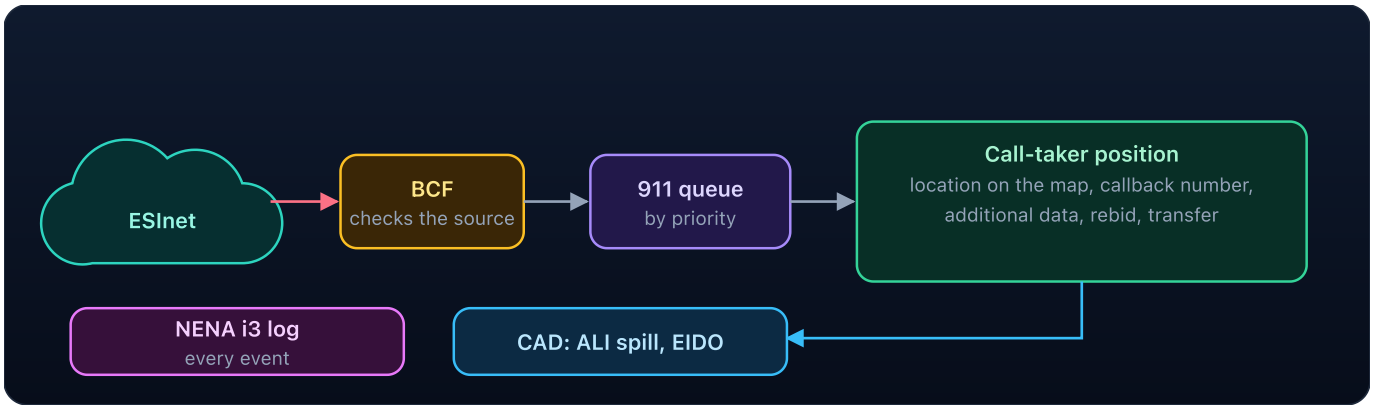


Figure 61. The 911 profile: an NG911 call reaches a call-taker

Setting up.

1. On Trunks (chapter 4), add the trunk from your ESInet provider (or the NG911 core's ESRP) and switch on **From the ESInet**.
2. In Call routing (chapter 5), send its calls to your call-takers: a queue, or a ring group.
3. On **911 calls**, enter your agency ID (a domain, like `psap.county.example.gov`) and what CAD takes: the address and port its ALI listener is on, and its EIDO address.

When a 911 call rings, it shows on **911 calls** with:

- **the address** the network sent (civic location), and **the position** with how sure it is (within so many metres), with a link to a map;
- **the provider**, the kind of service (wireless, VoIP, residence, business), the device and the subscriber, when the call brought them (Additional Data, RFC 7852);
- anything that went wrong reading it.

Location comes in the call (by value), or as a reference BitVoice fetches with HELD. A call with a reference has **Rebid location**: BitVoice asks again, for a caller who is moving or whose first fix was rough.

When a call-taker answers, CAD gets, each on its own so a CAD that is down never holds up the call:

- **the ALI spill**: the text block CAD systems read (callback number, class of service, address, position and uncertainty, provider), sent over TCP;
- **the EIDO**, the incident document of NENA-STA-024, by HTTPS POST (with a bearer token if set). **EIDO** on a call shows it.

The i3 log keeps the NENA i3 events of each call (call start, answer, location, EIDO sent), one file a day. **Answer times** shows how many calls were answered within your goal (NENA: 90% within 15 seconds, 95% within 20), the abandoned ones, and each call-taker's average.

What is not here yet: text to 911 (RTT and SMS), TTY, and transfer to another PSAP with the location carried along.

CHAPTER 62

Radio dispatch console

In the web interface: `/radio`

> **Lab and design-partner track.** Not certified for live 911 or radio dispatch, and not yet tried with a live ESInet, P25 system or CAD. Use it in a lab or in shadow mode beside your current equipment. Switch on **Public-safety profiles (lab)** in Settings (chapter 9) to show it in the menu.

Dispatchers talk on radio channels from the same headset as their phone calls. BitVoice reaches the radios through radio-over-IP (RoIP) gateways; the radio network itself stays with whoever runs it.

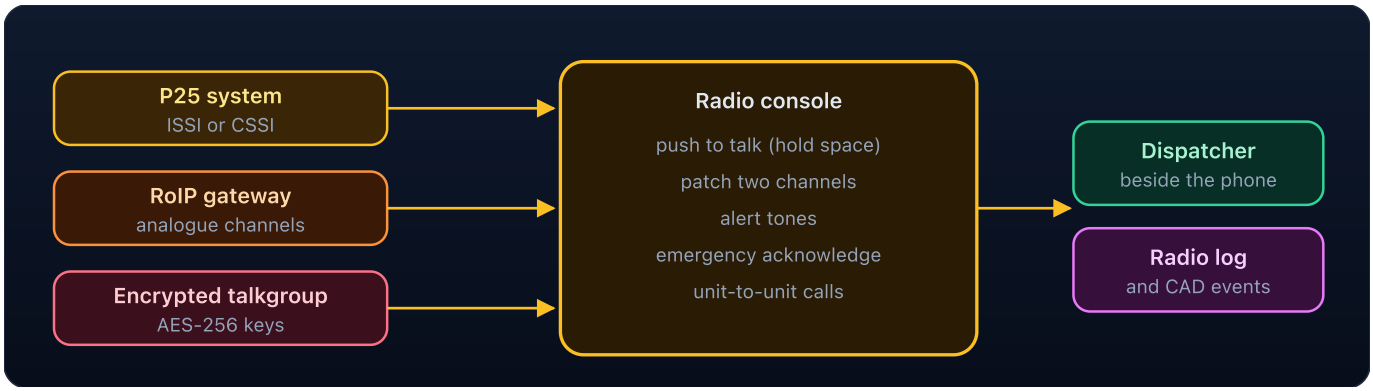


Figure 62. The radio console: talkgroups beside the phone

Setting up. Under **Console setup**, choose a console number (800, say) and add each channel: its name, the port on this server the gateway sends the channel's audio to, and the gateway's address and port for the audio going out. On the gateway: RTP, G.711 μ-law, to this server's address and that port; key the radio on voice (VOX). Each channel uses its port and the next one, outside the range phone calls use.

Joining. A dispatcher dials the console number from their desk phone, or presses **Join with the browser phone**. They then hear every channel's traffic mixed together; **Hearing / Muted** on a channel takes it out of their headset. The lamp on a channel lights while it is receiving.

Talking. Hold a channel's **Hold to talk** button (or focus it and hold the space bar): your voice goes to that channel only, and the channel shows **TX** with your extension. One dispatcher talks on a channel at a time; the button is greyed for the others while someone is talking. Nothing is sent to a channel while nobody is talking, so a voice-keyed radio is never held on the air.

Patching. Press **Patch** on one channel, then **Patch here** on another: the two hear each other (a fire channel and a mutual-aid channel, say) until you press **Unpatch**.

P25 talkgroups. A channel can also be a P25 talkgroup, joined directly over the ISSI (TIA-102.BACA, the interface between P25 subsystems; with **Send the CSSI console PTT word** on, BitVoice adds the console subsystem's word to each request). Choose **A P25 talkgroup (ISSI)**, give the talkgroup ID, the SIP address and ID of the RFSS that is home to the talkgroup, and a port for its voice. Under **This console on the P25 system**, enter the WACN and system ID, the RFSS ID your P25 administrator gave this console, the unit ID dispatchers transmit as, and the SIP port the ISSI uses here; the P25 system must know this server as an RFSS at that address. The tile shows **P25** while the group call is up (**P25 down** and the reason when it is not), and the unit ID of the radio talking, in red with EMERGENCY when it is an emergency call. **Hold to talk** asks the home RFSS for the channel (**Asking for the channel...**); your voice goes once it grants it, and if it refuses (someone with a higher priority has it) the tile says so. Voice is converted to and from IMBE, the P25 vocoder, in BitVoice.

Encrypted talkgroups. Under **Encryption keys**, add each traffic key your radio shop gives you: a name, the algorithm (AES-256, DES-OFB or triple DES), the key ID and the key itself in hex (TIA-102.AAAD). A key is never shown again once saved, not even to administrators, and leaving the field empty keeps it. The console hears an encrypted transmission when it has the key with that algorithm and key ID: the tile shows a lock and the algorithm next to the talking unit. Without the key the transmission is muted, never played as noise, and the tile says **no key: muted**. To transmit encrypted on a talkgroup, choose a key under **Transmit** for that channel; the tile then shows **TX AES-256** (or the algorithm you chose). A full configuration export (Settings, an administrator's backup) carries the keys, so keep those files as safe as the keys.

Private calls. With the console on a P25 system, **Call a radio** in your position's bar calls one radio by its unit ID (turn on **Emergency** to call it as an emergency). The call becomes a tile of its own, **Private call: unit 197**, with **Hold to talk** like any channel and **Hang up**. Radios can call the console too, on the unit ID dispatchers transmit as: the call appears as a tile marked **called in** (in red with **EMERGENCY** when the radio called as one), every dispatcher hears it, and it goes when either side hangs up. Under **Private calls** in the console's P25 settings, give the SIP address and ID of the RFSS that is home to the radios (empty uses the first talkgroup's home) and, if private calls should be encrypted, the key to send them with.

Alert tones. Send an alert tone... on a channel sends one of: **Alert 1**, a steady 1000 Hz tone for 3 seconds; **Alert 2**, a warble between 1000 and 1500 Hz; **Alert 3**, 1000 Hz pulses; or a short **Beep**. The tone holds the channel while it plays (the tile shows **tone** and your extension), every dispatcher monitoring the channel hears it go out, and on a P25 talkgroup it goes once the home RFSS grants the channel.

Emergencies. When a radio transmits as an emergency (its emergency button), its tile turns red with **EMERGENCY** and the unit ID, the server log notes it, and CAD is told at once (below).

Radio log and recording. Turn on **Record every transmission** under **Recording and CAD**: each transmission on each channel, heard or sent, is kept as a recording with a line in the **Radio log**: when, the channel, who (the radio's unit, or the dispatcher's extension), how long, and whether it was an

emergency, an alert tone, encrypted or a private call. A heard transmission starts with the first speech and ends 1.5 seconds after the last word. Find a unit, dispatcher or channel with the search box, and press **Play** to listen. Recordings are kept as long as call recordings are (Settings, Recording).

Radio data to CAD. Give CAD's address under **Send radio events to CAD at** (and a shared secret if CAD wants one; it is sent as a bearer token and never shown again). BitVoice then POSTs each event as JSON, `{"type": ..., "agency": ..., "data": {...}}`: **radio.unit** when a radio starts talking on a channel (its unit ID, WACN, system, the channel, emergency, encryption), **radio.emergency** for an emergency (tried three times if CAD does not answer), and **radio.transmission** for each finished transmission as it appears in the radio log. CAD can also read the log itself with `GET /api/radio/log` (filters: `channel`, `q`, `since`, `until`). A CAD that is down never holds up the console. P25 radios' GPS positions are not part of this: they travel over P25 packet data (location services), not over the ISSI's voice.

Not here yet: P25 Phase 2 (TDMA) voice in its own half-rate vocoder. A Phase 2 system works with the console when its ISSI talkgroups carry full-rate (Phase 1) voice or its transcoder converts them, which is common: ask your P25 administrator. A system sending native half-rate voice over the ISSI does not work yet; the half-rate vocoder (AMBE+2) is patented until 2028 and needs a licence from its owner or a hardware vocoder chip.

CHAPTER 63

NG911 Core Services (ECRF, ESRP, LVF, BCF)

In the web interface: `/core`

> **Lab and design-partner track.** Not certified for live 911 or radio dispatch, and not yet tried with a live ESInet, P25 system or CAD. Use it in a lab or in shadow mode beside your current equipment. Switch on **Public-safety profiles (lab)** in Settings (chapter 9) to show it in the menu.

For a state or regional 911 authority that runs its own NG911 core: BitVoice answers LoST as the ECRF, and routes each 911 call as the ESRP to the PSAP that serves the caller.

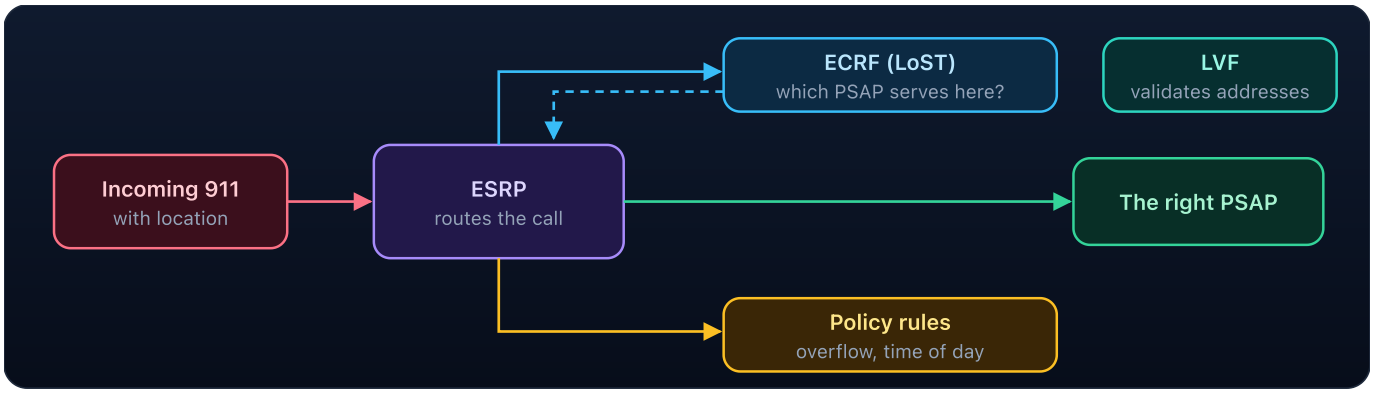


Figure 63. NG911 Core Services: routing a call by where the caller is

Service areas. Export your boundaries from GIS as GeoJSON polygons (PSAP boundaries for `urn:service:sos`; police, fire or EMS areas for `urn:service:sos.police` and the others) and upload them on **Core services**. A polygon's `name`, `uri` and `number` properties are read; give a default PSAP URI for those without one. **Edit** an area to add the cities or counties it serves, for callers whose location is a civic address only. **Try a place** shows which area answers a position or a city, as a LoST query would.

The ECRF. Switch on **Answer LoST queries** and list the addresses that may ask (your ESInet): other elements send findService to `/lost` (RFC 5222) and get the PSAP's URI. A specific service (sos.police) is answered from its own areas first, then from sos.

The ESRP. List the PSAPs: each one's URI (as the areas name it), the trunk that reaches it and its overflow PSAP, and choose the default PSAP. Then send the ESInet's incoming 911 calls to **ESRP** in Call routing (chapter 5). Each call goes to the PSAP for its location (from this server's areas, or an **outside ECRF** by LoST), with its location and the service it asked for carried on; when that PSAP does not take it, to its overflow; without a location, or from a place no area covers, to the default. Each decision is in the i3 log on 911 calls (chapter 61).

Policy rules. Once the ECRF has picked a PSAP, the rules under **Policy rules** are checked in order and the first whose conditions all hold decides: send the call to another PSAP, or refuse it as busy. A rule can look at the service (fire calls only, say), the PSAP the ECRF chose, a schedule (while the night shift's centre is closed), calls with no location, calls the BCF marked suspicious, and the trunk the call came in on. Each decision is in the i3 log as a RouteLogEvent naming the rule.

Border control (BCF). With **Check calls from the ESInet** on, every call from an ESInet trunk is checked first: addresses under **Always refuse** are turned away, and so is an address that starts more calls a minute, or holds more calls at once, than you allow (a telephony denial-of-service attack sends floods of calls from a few sources). Refused calls get 503 with Retry-After. Turn on **let the call through marked suspicious** to send such calls on instead and let a policy rule decide (to a centre that screens them, or refused); a real caller in a flood is then not lost. **Border control: latest** lists what was refused or marked.

Address validation (LVF). Upload your 911 GIS address data as GeoJSON, in the NENA GIS data model (NENA-STA-006): road centerlines with `FromAddr_L`, `ToAddr_L`, `Parity_L`, `IncMuni_L` and the rest,

and site/structure address points with `Add_Number`, `St_Name`, `Inc_Muni`. A LoST query with `validateLocation="true"` then gets the address's valid, invalid and unchecked parts in its answer (RFC 5222), so carriers and location servers can correct an address before a 911 call ever uses it. **Check an address** does the same here: an exact address point is the best match, then a street segment whose range on the right side of the road holds the number (odd and even sides respected); spellings like Street and St, North and N are treated alike.

Not here yet: the BCF's media checks (it limits and denies by address; it does not inspect SIP or media for attacks), similar-address suggestions for invalid ones, and spatial checks of a position against the address data.

BitVoice is a product of BioDream LLC, published under the Apache License 2.0. The same chapters are under **Help** in every BitVoice.